

Preface

At the root of measure theory is a rigorous treatment of how to extend the notions of length (on the real line), area (on the Euclidean plane), and volume (in three-space) to a larger collection of sets than those seen in, say, courses on Calculus. We have seen in a first course in analysis that there exist sets with some exceptionally strange properties (the Cantor set being a classic example). We would like to be able to discuss these types of sets in the context of measuring their sizes (be it a length, area, volume, etc). While the historical impetus was to solve this problem for sets in $\mathbb{R}^n$, one finds that it is not hard to abstract the theory to more general sets. Accordingly we take up this abstract study. This is not just for the sake of abstraction. In fact one can find applications for abstract measure theory in, for example, probability theory (where the measure of a set corresponds to the likelihood of an element in that set being chosen under some random process) and physics (where one might want the measure of a set to correspond to the mass of some object occupying that space).

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Chapter 1

Preliminaries

§ 1.1. Set Theory

Definition 1.1.1. A *set* is a collection of different things; the things are called *elements* of the set.

Zermelo–Fraenkel set theory introduces eight different axioms which mathematics builds upon. Below is an introduction to the ones which are relevant for the material covered in this course.

Axiom 1 (Axiom of Extension). *If the sets X and Y have the same members, then they are the same set. In other words:*

$$(\forall X)(\forall Y)\left((\forall z)(z \in X \Leftrightarrow z \in Y) \implies X = Y\right)$$

Axiom 2 (Axiom of Existence). *There is a set such that no element is a member of it. In other words:*

$$(\exists A) : (\forall x)(x \notin A)$$

Note that the Axiom of Extension immediately proves that the set which contains no elements is unique; two sets are equal if they have the same members, hence two sets which contain no elements are equal.

Definition 1.1.2. The set which contains no elements is called the *empty set*, and is denoted by $\emptyset$.

Axiom 3 (Axiom of Pairing). *Let A and B be sets. There exists the set $\mathcal{C} := \{A, B\}$. In other words:*

$$(\forall A)(\forall B)(\exists \mathcal{C}) : (A \in \mathcal{C} \wedge B \in \mathcal{C}).$$

Axiom 4 (Axiom of Union). *Let $\mathcal{C}$ be a set. Then there exists a set $\mathcal{U}$ consisting of the elements x that are contained in at least one element of the set $\mathcal{C}$, that is:*

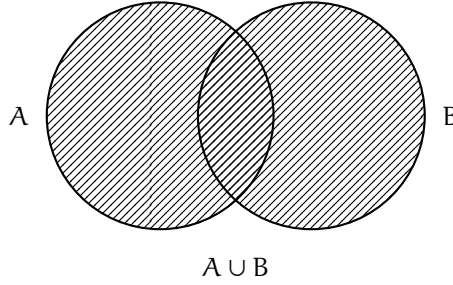
$$(\forall \mathcal{C})(\exists \mathcal{U}) : (\forall x)(x \in \mathcal{U} \Leftrightarrow (\exists C)(x \in C \wedge C \in \mathcal{C}))$$

Definition 1.1.3. Let $\mathcal{C}$ be a set, and let $\mathcal{U}$ be the set consisting of the elements x that are contained in at least one element of $\mathcal{C}$.

(1) The set U is called the *union of the set $\mathcal{C}$* and is denoted:

$$U := \bigcup \mathcal{C}.$$

(2) If the set $\mathcal{C}$ consists of only two sets A and B , then we write $U := A \cup B$.



Axiom 5 (Axiom Schema of Specification). *Let A be a set, and let φ be a predicate containing the free variable x . There exists a subset B of the set A whose members are precisely the elements x of the set A such that the sentence $\varphi(x)$ is true. In other words:*

$$(\forall A)(\forall \varphi)(\exists B) : (B = \{x \in A \mid \varphi(x)\})$$

Proposition 1.1.1. *Let $\mathcal{C}$ be a nonempty set. There exists a set D consisting of the elements x that are contained in all elements of the set $\mathcal{C}$.*

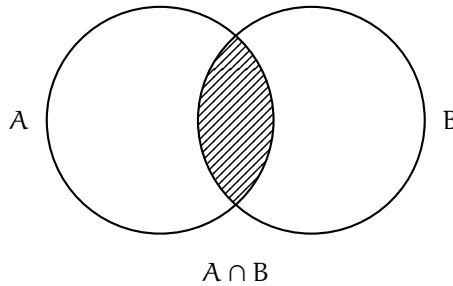
Proof. Let $E \in \mathcal{C}$. By the Axiom Schema of Specification we have $D := \{c \in E \mid (\forall X)(X \in \mathcal{C} \Rightarrow c \in X)\}$. $\square$

Definition 1.1.4. Let $\mathcal{C}$ be a nonempty set, and let D be the set consisting of the elements x that are contained in all elements of the set $\mathcal{C}$.

(1) The set D is called the *intersection of the set $\mathcal{C}$* and is denoted:

$$D := \bigcap \mathcal{C}.$$

(2) If the set $\mathcal{C}$ contains only two sets A and B , then we write $D := A \cap B$.



Axiom 6. *Let X be a set. Then the set of all subsets of X exists.*

Definition 1.1.5. Let X be a set. Then the set of all subsets of the set X is called the *power set* of the set X and is denoted $\mathcal{P}(X)$.

Axiom 7 (Axiom of Infinity). *There is a set that contains all natural numbers.*

Definition 1.1.6. An *index set* is a nonempty set whose elements label the elements of another set. We denote the collection of objects labeled by elements of I as $\{A_\alpha\}_{\alpha \in I}$.

With this definition and the Axiom of Infinity, we can extend the concept of unions and intersections to families of sets of arbitrary size.***

Definition 1.1.7. Let I be an index set, let $\{X_\alpha\}_{\alpha \in I}$ be an indexed family of sets, and write $\mathcal{X} := \{X_\alpha \mid \alpha \in I\}$. Define:

$$\bigcup_{\alpha \in I} X_\alpha := \bigcup \mathcal{X},$$

$$\bigcap_{\alpha \in I} X_\alpha := \bigcap \mathcal{X}.$$

Definition 1.1.8. *** Let $\{X_n\}_{n \in \mathbb{N}}$ be a family of sets indexed by the natural numbers.

(1) The *limit superior* of $\{X_n\}_{n \in \mathbb{N}}$ is:

$$\begin{aligned} \limsup X_n &:= \bigcap_{k=1}^{\infty} \left(\bigcup_{n=k}^{\infty} X_n \right) \\ &= \{x \mid x \in E_n \text{ for infinitely many } n\}. \end{aligned}$$

(2) The *limit inferior* of $\{X_n\}_{n \in \mathbb{N}}$ is:

$$\begin{aligned} \limsup X_n &:= \bigcup_{k=1}^{\infty} \left(\bigcap_{n=k}^{\infty} X_n \right) \\ &= \{x \mid x \notin E_n \text{ for only finitely many } n\} \end{aligned}$$

Definition 1.1.9. Let A and B be two sets. Define the *difference* of A and B by $A \setminus B := \{x \in A \mid x \notin B\}$.

Note that the difference of two sets exists by the Axiom Schema of Specification.

Definition 1.1.10. Let a and b be two objects. The *ordered pair* (a, b) is defined by $(a, b) := \{\{a\}, \{a, b\}\}$.

Proposition 1.1.2. Let A and B be sets. There exists a set E containing all ordered pairs (a, b) , where $a \in A$ and $b \in B$.

Proof. Let (a, b) be any ordered pair such that $a \in A$ and $b \in B$. By definition $(a, b) = \{\{a\}, \{a, b\}\}$. Now $\{a\} \subset A$, so $\{a\} \subset A \cup B$, hence $\{a\} \in \mathcal{P}(A \cup B)$. Similarly, $\{a, b\} \subset A \cup B$, so $\{a, b\} \in \mathcal{P}(A \cup B)$. It follows that $\{\{a\}, \{a, b\}\} \subset \mathcal{P}(A \cup B)$, hence $\{\{a\}, \{a, b\}\} \in$

$\mathcal{P}(\mathcal{P}(A \cup B))$. We know all these sets are known to exist by the Axiom of Union and Axiom of Power Set. Then by the Axiom Schema of Specification, there exists a set:

$$E := \{t \in \mathcal{P}(\mathcal{P}(A \cup B)) \mid (\exists a)(\exists b)(a \in A \wedge b \in B \wedge t = (a, b))\}.$$

Moreover, this set is unique by the Axiom of Extension, and is the set of all ordered pairs (a, b) with $a \in A$ and $b \in B$. $\square$

Definition 1.1.11. Let A and B be two sets, and let E be the set containing all ordered pairs (a, b) where $a \in A$ and $b \in B$. The set E is called the *Cartesian product of A and B* , and is denoted $E := A \times B = \{(a, b) \mid a \in A \wedge b \in B\}$.

Definition 1.1.12. Let X and Y be sets. A *relation* from X to Y is a subset R of $X \times Y$. We write xRy to mean $(x, y) \in R$.

- (1) An *equivalence relation* on X is a relation which is reflexive, transitive, and symmetric. We write $x \sim y$ to mean xRy .
- (2) A *function* (or *map*) from X to Y , denoted $f : X \rightarrow Y$, is a relation from X to Y such that, for every $x \in X$, there exists a unique $y \in Y$ such that xRy . We write $f(x) = y$ to mean xRy .
- (3) A *partial order* on X is a relation which is reflexive, transitive, and antisymmetric. We write $x \leq y$ to mean xRy , and say that X is an *ordered set*.

Proposition 1.1.3. *** Let X be a set. The following are equivalent:

- (1) S is a partition of X ;
- (2) There exists an equivalence relation $\sim$ on X such that the equivalence classes of $\sim$ are exactly the elements of S .

Proof. $\square$

Example 1.1.1. An equivalent relation on X partitions X into equivalence classes.

Example 1.1.2. Recall that, if $\mathcal{C}$ and $\mathcal{D}$ are categories, then a *covariant functor* $T : \mathcal{C} \rightarrow \mathcal{D}$ is a map such that:

- (i) If $A \in \text{obj}(\mathcal{C})$, then $T(A) \in \text{obj}(\mathcal{D})$.
- (ii) If $f \in \text{Hom}_{\mathcal{C}}(A, A')$, then $T(f) \in \text{Hom}_{\mathcal{D}}(T(A), T(A'))$.
- (iii) If $f \in \text{Hom}_{\mathcal{C}}(A, A')$ and $g \in \text{Hom}_{\mathcal{C}}(A', A'')$, then $T(f) \in \text{Hom}_{\mathcal{D}}(T(A), T(A'))$ and $T(g) \in \text{Hom}_{\mathcal{D}}(T(A'), T(A''))$. In particular:

$$T(gf) = T(g)T(f).$$

- (iv) For every $A \in \text{obj}(\mathcal{C})$, $T(\text{id}_A) = \text{id}_{T(A)}$.

A *contravariant functor* $T : \mathcal{C} \rightarrow \mathcal{D}$ is similarly defined, but (ii) and (iii) are instead defined as:

- (ii) If $f \in \text{Hom}_{\mathcal{C}}(A, A')$, then $T(f) \in \text{Hom}_{\mathcal{D}}(T(A'), T(A))$.

- (iii) If $f \in \text{Hom}_{\mathcal{C}}(A, A')$ and $g \in \text{Hom}_{\mathcal{C}}(A', A'')$, then $T(f) \in \text{Hom}_{\mathcal{D}}(T(A'), T(A))$ and $T(g) \in \text{Hom}_{\mathcal{D}}(T(A''), T(A'))$. In particular:

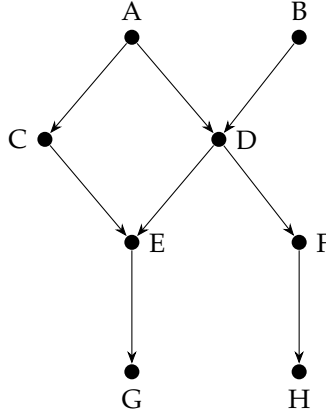
$$T(gf) = T(f)T(g).$$

Given $f : X \rightarrow Y$, the power set gives rise to two functors, the *contravariant power set functor* $\mathbf{Set}^{\text{op}} \rightarrow \mathbf{Set}$ and the *covariant power set functor* $\mathbf{Set} \rightarrow \mathbf{Set}$. The first sends $f : X \rightarrow Y$ to the *preimage* function $f^{-1} : \mathcal{P}(Y) \rightarrow \mathcal{P}(X)$, whereas the second sends f to the *image* function $f_* : \mathcal{P}(X) \rightarrow \mathcal{P}(Y)$. The preimage is more well-behaved, commuting with boolean operations:

$$\begin{aligned} f^{-1}(E^c) &= (f^{-1}(E))^c, \\ f^{-1}\left(\bigcup_{\alpha \in I} E_{\alpha}\right) &= \bigcup_{\alpha \in I} f^{-1}(E_{\alpha}), \\ f^{-1}\left(\bigcap_{\alpha \in I} E_{\alpha}\right) &= \bigcap_{\alpha \in I} f^{-1}(E_{\alpha}). \end{aligned}$$

This is not necessarily the case for the image function.

Example 1.1.3. *****(I don't like the prose here) Consider the directed acyclic graph:



Let $G = \{A, B, C, D, E, F, G, H\}$ be the set of vertices of our graph. We can define a partial ordering on G as follows: for any $x, y \in G$, we have $x \leq y$ if and only if there exists a directed path from x to y . However, note that not every element of G can be compared.

Definition 1.1.13. An ordering on a set X is said to be *linear* (or *total*) if for every $x, y \in X$, $x \leq y$ or $y \leq x$.

Definition 1.1.14. Let X be an ordered set. Let $A \subset X$.

- (1) A is called *bounded above* if there exists an element $u \in X$ with $a \leq u$ for all $a \in A$. Such a u is called an *upper bound* for A . The set of upper bounds of A is denoted $\mathcal{U}_A = \{u \in X \mid u \text{ is an upper bound of } A\}$.

- (2) A is called *bounded below* if there exists an element $v \in X$ with $v \leq a$ for all $a \in A$. Such a v is called a *lower bound* for A . The set of lower bounds of A is defined as $\mathcal{L}_A = \{v \in X \mid v \text{ is a lower bound of } A\}$.
- (3) If A admits an upper bound u with $u \in A$, then u is called *the greatest element of } A.*
- (4) If A admits a lower bound v with $v \in A$, then v is called *the least element of } A.*
- (5) If l is the least element of $\mathcal{U}_A$, we write $l = \sup(A)$ and call it the *supremum of } A.*
- (6) If g is the greatest element of $\mathcal{L}_A$, we write $g = \inf(A)$ and call it the *infimum of } A.*
- (7) A *maximal element of } A* is an element $m \in A$ such that if $a \geq m$, then $a = m$ (not necessarily unique).
- (8) A *minimal element of } A* is an element $n \in A$ such that if $a \leq n$, then $a = n$ (not necessarily unique).

Definition 1.1.15. If a linear order on a set X satisfies the property that every nonempty subset has a minimal element, then we say it is a *well-ordering*.

—————//—————

***include stuff HERE

Definition 1.1.16.

do general cartesian product
 question is it nonempty?
 axiom of choice
 answer: no, axiom of choice says this choice function exists.
 'shoes vs. socks' formulation of the axiom

—————//—————

§ 1.2. The Extended Real Numbers***

Definition 1.2.1. The *extended real number line* is the set $\overline{\mathbb{R}} := \mathbb{R} \cup \{\infty, \infty\} = [-\infty, \infty]$.

The extended real number line has many unique properties.

- For all $x \in \mathbb{R}$, we have $-\infty < x < \infty$. This allows us to extend the ordering of $\mathbb{R}$ to $\overline{\mathbb{R}}$.
- Recall that since $\mathbb{R}$ is complete, every *bounded subset* admits a supremum. However, for the extended real number line, *every subset* of $\overline{\mathbb{R}}$ admits a supremum.

Example 1.2.1.

- (i) In $\overline{\mathbb{R}}$, we have $\sup \mathbb{R} = \infty$.
- (ii) $\inf \mathbb{Q} \cap (-\infty, 0) = -\infty$.

(iii) $\inf \emptyset = \infty$. This follows from basic intuition: the infimum of a set is the greatest lower bound, so the infimum of the empty set is an element which is a lower bound for nothing. Hence it must be ∞ .

- In $\mathbb{R}$, any sequence which is monotone and bounded converges. However, in $\overline{\mathbb{R}}$, *any* monotone sequence converges to a value in $\overline{\mathbb{R}}$.
- For any sequence $(x_n)_n$, we define:

$$\limsup x_n := \inf_{k \geq 1} \left(\sup_{n \geq k} x_n \right),$$

$$\liminf x_n := \sup_{k \geq 1} \left(\inf_{n \geq k} x_n \right).$$

In $\mathbb{R}$, it could be the case that each is properly divergent, however in $\overline{\mathbb{R}}$ the limit superior and limit inferior *always* exist.

- We can define arithmetic with $-\infty$ and ∞ . For $x \in \overline{\mathbb{R}}$:

$$\begin{aligned} x + \infty &= \infty, \\ x - \infty &= -\infty, \\ \infty + \infty &= \infty, \\ -\infty - \infty &= -\infty. \end{aligned}$$

We don't give any meaning to $\infty - \infty$. If $x > 0$, we define:

$$\begin{aligned} x \cdot \infty &= \infty, \\ x \cdot (-\infty) &= -\infty. \end{aligned}$$

If $x < 0$, we define:

$$\begin{aligned} x \cdot \infty &= -\infty, \\ x \cdot (-\infty) &= \infty. \end{aligned}$$

Unless stated otherwise, we define $0 \cdot \pm\infty = 0$.

- Let X be any set and let $f : X \rightarrow [0, \infty]$. We'd like to give some meaning to $\sum_{x \in X} f(x)$. Regardless of the cardinality of X , it is a theorem that

This whole section needs to be redone.

summations are defined like this: $\sum_{x \in X} f(x) = \sup \{ \sum_{x \in Y} f(x) \mid Y \subset X, Y \text{ finite} \}$.

Proposition 1.2.1. **** (could be cleaned up) Every open set in $\mathbb{R}$ is at most a countable union of disjoint open intervals.*

Proof. Let $U \subset \mathbb{R}$ be open. Define a binary relation $\sim$ on U as follows: $x \sim y$ if and only if there exists an open interval I such that $\{x, y\} \subset I \subset U$. One can check that this is indeed an equivalence relation, hence U is partitioned by the equivalence classes of $\sim$.

We are now going to show that each equivalence class is an open interval. Let E be an equivalence class and let $I = (\inf E, \sup E)$. Clearly $E \subset I$. Let $x \in I$. Then $\inf E < x$ and $x < \sup E$. We can find $u, v \in E$ such that $\inf E < u < x < v < \sup E$. Since $u, v \in E$, we can find an interval J such that $\{u, v\} \subset J \subset U$. But this means $x \in J$, hence $x \sim u$ and $x \sim v$. Thus $x \in E$, giving $I \subset E$ meaning that E is an open interval.

It remains to show that there are at most a countable number of equivalence classes. Note that each equivalence class contains a rational number, and no two classes can contain the same rational number (otherwise the equivalence classes wouldn't be disjoint). By the Axiom of Choice, we can find a choice function $f : U/\sim \rightarrow \mathbb{Q}$ which is injective. Therefore there are at most countably many open intervals. $\square$

§ 1.3. Metric Spaces

Definition 1.3.1. A *distance function* (or *metric*) on a nonempty set X is a map:

$$\rho : X \times X \rightarrow [0, \infty)$$

satisfying for all $x, y, z \in X$:

- (1) $\rho(x, x) = 0$;
- (2) $\rho(x, y) = \rho(y, x)$;
- (3) $\rho(x, z) \leq \rho(x, y) + \rho(y, z)$.

The pair (X, ρ) is called a *metric space*.

Distance functions let us talk about sets which centered at a point.

Definition 1.3.2. Let X be a metric space, $x \in X$ and $r > 0$.

- (1) The *open ball of radius r centered at x* is the set $B(r, x) := \{y \in X \mid \rho(x, y) < r\}$.
- (2) The *closed ball of radius r centered x* is the set $V(r, x) := \{y \in X \mid \rho(x, y) \leq r\}$.
- (3) The *sphere of radius r centered at x* is the set $S(r, x) := \{y \in X \mid \rho(x, y) = r\}$.

Such distance functions induce a topology on our metric space X .

Definition 1.3.3. Let X be a metric space.

- (1) We say $U \subset X$ is *open* if, for all $x \in U$, there exists $r > 0$ such that $B(r, x) \subset U$.
- (2) We say $V \subset X$ is *closed* if V^c is open.

Given a set, we characterize its inside, the smallest closed set containing it, as well as its boundary.

Definition 1.3.4. Let X be a metric space and $E \subset X$.

- (1) The *interior of E* is the set:

$$E^\circ := \bigcup \{U \subset E \mid U \text{ is open}\}.$$

(2) The *closure* of E is the set:

$$\bar{E} := \bigcap \{C \supset E \mid C \text{ is closed}\}.$$

(3) The *boundary* of E is the set:

$$\partial E = \bar{E} \setminus E^\circ.$$

Proposition 1.3.1. *A countable subset of $\mathbb{R}$ has an empty interior.*

Proof. Let A be a subset of $\mathbb{R}$ which has a nonempty interior. Then A contains a non-trivial interval. Hence $\text{card}(A) \geq \aleph_0$. Thus A is uncountable. $\square$

Example 1.3.1. Let $E = \mathbb{Q} \cap (0, 1)$. We have $E^\circ = \emptyset$ and $\bar{E} = [0, 1]$.

Definition 1.3.5. Let X be a metric space and $E \subset X$.

- (1) E is *dense* in X if $\bar{E} = X$.
- (2) E is *nowhere dense* if $(\bar{E})^\circ = \emptyset$.

Intuitively, a set is nowhere dense if we can't fit an open ball inside it.

Definition 1.3.6. Let (X, d) be a metric space.

- (1) The collection of open sets of X is denoted τ_X .
- (2) A *base* for τ_X is a family of open subsets $\mathcal{B} \subseteq \tau_X$ such that:

$$(\forall U \in \tau_X)(\forall x \in U)(\exists B \in \mathcal{B}) : x \in B \subseteq U.$$

Equivalently, for all $U \in \tau_X$, we can write $U = \bigcup_{i \in I} B_i$, where $\{B_i\}_{i \in I} \subseteq \mathcal{B}$.

- (3) X is *second countable* if it has a countable base.

Definition 1.3.7. Let (X, ρ) be a metric space. X is *separable* if there exists a countable dense subset.

Proposition 1.3.2. *** *Let (X, ρ) be a metric space. X is second countable if and only if X is separable.*

Proof. $\square$

With a metric we can talk about the distance from a point to a set, as well as a way to define if a set is bounded.

Definition 1.3.8. Let (X, ρ) be a metric space, $x \in X$, and $E \subset X$.

- (1) The distance from x to E is $\rho(x, E) := \inf_{y \in E} \rho(x, y)$.
- (2) The *diameter* of E is $\text{diam}(E) = \sup_{x, y \in E} \rho(x, y)$.
- (3) A subset of a metric space is *bounded* if its diameter is finite.

With a notion of distance, we can talk about sequence convergence.

Definition 1.3.9. Let (X, ρ) be a metric space and $(x_n)_n$ a sequence in X . We say $(x_n)_n$ converges to a point x_0 if $\lim_{n \rightarrow \infty} \rho(x_n, x_0) = 0$.

Proposition 1.3.3. Let (X, ρ) be a metric space, $E \subset X$, and $x \in X$. The following are equivalent:

- (1) $x \in \bar{E}$;
- (2) For every $r > 0$, $B(r, x) \cap E \neq \emptyset$;
- (3) There is a sequence $(x_n)_n$ in E which converges to x

We can even talk about continuity of functions between metric spaces.

Definition 1.3.10. Let (X_1, ρ_1) and (X_2, ρ_2) be metric spaces and $f : X_1 \rightarrow X_2$.

- (1) We say f is *continuous* if:

$$(\forall c \in X_1)(\forall \epsilon > 0)(\exists \delta > 0) : (\forall x \in X_1)(\rho_1(x, c) < \delta \implies \rho_2(f(x), f(c)) < \epsilon).$$

- (2) We say f is *uniformly continuous* if:

$$(\forall \epsilon > 0)(\exists \delta > 0) : (\forall x, y \in X_1)(\rho_1(x, y) < \delta \implies \rho_2(f(x), f(y)) < \epsilon).$$

- (3) We say f is *Lipschitz* if:

$$(\exists C \geq 0) : (\forall x, y \in X_1)(\rho_2(f(x), f(y)) \leq C \cdot \rho_1(x, y)).$$

Note that δ may depend on our point c in the case that f is continuous. A function is uniformly continuous if δ does not depend on any two points in its domain.

Proposition 1.3.4. Let (X_1, ρ_1) and (X_2, ρ_2) be metric spaces. f is continuous if and only if the preimage of every open set in X_2 is open in X_1 .

Similarly to how every Cauchy sequence in $\mathbb{R}$ converges, we can talk about metric spaces which guarantee that same property.

Definition 1.3.11.

- (1) Let $(x_n)_n$ be a sequence in a metric space (X, ρ) . We say $(x_n)_n$ is ρ -Cauchy if:

$$(\forall \epsilon > 0)(\exists N \in \mathbb{N}) : (\forall n, m \in \mathbb{N})(n > m \geq N \implies \rho(x_n, x_m) < \epsilon).$$

- (2) A metric space X is called *complete* if every Cauchy sequence converges to a point in X .

We can generalize the notion of a set being closed and bounded in $\mathbb{R}^n$.

Proposition 1.3.5. Let X be a metric space. The following are equivalent:

- (1) Every open cover of X admits a finite subcover;
- (2) Every sequence in X admits a convergent subsequence;
- (3) X is complete and totally bounded.

Definition 1.3.12. We say a metric space X is *compact* if it satisfies any of the properties in Proposition 1.3.5.

A set may have more than one metric defined on it.

Definition 1.3.13. Let X be a metric space. We say ρ_1 and ρ_2 are *equivalent metrics on X* if $\text{id} : (X, \rho_1) \rightarrow (X, \rho_2)$ and $\text{id}^{-1} : (X, \rho_2) \rightarrow (X, \rho_1)$ are Lipschitz.

Chapter 2

Measures

Ideally we would like to have a function μ that assigns each $E \subset \mathbb{R}$ to a number $\mu(E) \in [0, \infty]$. Such a function μ should possess the following properties:

1. The unit interval should have a measure equal to one—that is, $\mu([0, 1)) = 1$;
2. If E and F are congruent subsets of $\mathbb{R}$ then $\mu(E) = \mu(F)$;
3. If $E_1, E_2, \dots$ are pairwise disjoint, then $\mu\left(\bigcup_{j=1}^{\infty} E_j\right) = \sum_{j=1}^{\infty} \mu(E_j)$.

Unfortunately, a function possessing all of these traits does not exist.

Theorem 2.0.1. *A function $\mu : \mathcal{P}(\mathbb{R}) \rightarrow [0, \infty]$ satisfying (1), (2), and (3) from above does not exist.*

Proof. Suppose such a function μ does exist. By (1) we have $\mu([0, 1)) = 1$. Define a relation on $[0, 1)$ as follows: $x \sim y$ if and only if $x - y \in \mathbb{Q}$. This is an equivalence relation. Using the Axiom of Choice, let N be a set which contains exactly one element from each equivalence class. Now for each $q \in \mathbb{Q} \cap [0, 1)$, define:

$$N_q = \{x + q \mid x \in N \cap [0, 1 - q)\} \cup \{x + (q - 1) \mid x \in N \cap [1 - q, 1)\}.$$

That is, the points in N_q are obtained by "rotating" the points in N by a rational number q . Clearly $N_q \subset [0, 1)$, hence $\bigcup_{q \in \mathbb{Q} \cap [0, 1)} N_q \subset [0, 1)$. We have the following three properties:

- (a) For any $q \in \mathbb{Q}$, $\mu(N) = \mu(N_q)$
- (b) If $r, s \in \mathbb{Q}$, $r \neq s$, then $N_r \cap N_s = \emptyset$.
- (c) For every $y \in [0, 1)$, there exists a $q \in \mathbb{Q}$ such that $y \in N_q$.

Property (a) follows from the fact that N and N_r are congruent, hence:

$$\begin{aligned} \mu(N) &= \mu(N \cap [0, 1 - q)) + \mu(N \cap [1 - q, 1)) \\ &= \mu(\{x + q \mid x \in N \cap [0, 1 - q)\}) + \mu(\{x + q - 1 \mid x \in N \cap [1 - q, 1)\}) \quad \text{"Rotation"} \\ &= \mu(N_r). \end{aligned}$$

Property (b) follows by contradiction—if $x \in N_r \cap N_s$, then $x - r$ and $x - s$ would be distinct elements of N . However, $(x - r) - (x - s) = s - r \in \mathbb{Q}$, so $x - r$ and $x - s$ are in the same equivalence class. But this contradicts the way which we've constructed N . For property (c), given $y \in [0, 1)$, let $x \in N$ satisfy $x \sim y$. Since x and y are in the same equivalence class, we can find $q \in \mathbb{Q}$ such that:

$$y = \begin{cases} x + q, & x < y \\ x - (q - 1) & x > y \end{cases}.$$

Hence $y \in N_q$, giving $[0, 1) \subset N_q \subset \bigcup_{q \in \mathbb{Q} \cap [0, 1)} N_q$. We've obtained the equality:

$$\bigcup_{q \in \mathbb{Q} \cap [0, 1)} N_q = [0, 1).$$

Property (b) says that this is a disjoint union:

$$\bigsqcup_{q \in \mathbb{Q} \cap [0, 1)} N_q = [0, 1).$$

Finally, with property (a), we have:

$$\begin{aligned} 1 &= \mu([0, 1)) \\ &= \mu\left(\bigsqcup_{q \in \mathbb{Q} \cap [0, 1)} N_q\right) \\ &= \sum_{q \in \mathbb{Q} \cap [0, 1)} \mu(N_q) \\ &= \sum_{n=1}^{\infty} \mu(N). \end{aligned}$$

This is impossible, hence there does not exist a μ satisfying (1), (2), and (3). $\square$

A good guess for this failure would be that we assumed countable additivity as opposed to finite additivity. This is not the case. In dimensions $n \geq 3$, assuming such a weak form of μ leads to the following result:

Theorem 2.0.2 (Banach-Tarski). *Let U and V be arbitrary bounded open sets in $\mathbb{R}^n$ for $n \geq 3$. There exist $k \in \mathbb{N}$ and subsets $E_1, \dots, E_k, F_1, \dots, F_k$ of $\mathbb{R}^n$ such that:*

- the E_j 's are disjoint and their union is U ;
- the F_j 's are disjoint and their union is V ;
- E_j is congruent to F_j for all $1 \leq j \leq k$.

Ultimately, the problem comes from the fact we defined μ on *every possible* subset of $\mathbb{R}$. We can't hope to do this, since as we showed there exists bizarre sets like N which cannot be measured. Instead, we will define μ on a special class of subsets of $\mathbb{R}^n$.

§ 2.1. σ -Algebras

Definition 2.1.1. Let X be a set. An *algebra of sets* $\mathcal{A} \subset \mathcal{P}(X)$ is a collection of subsets of X which is:

- (1) closed under finite union: if $E_1, E_2, \dots, E_n \in \mathcal{A}$, then $\bigcup_{i=1}^n E_i \in \mathcal{A}$;
- (2) closed under complement: if $E \in \mathcal{A}$, then $E^c \in \mathcal{A}$.

If $\mathcal{A}$ is closed under countable union, then we say $\mathcal{A}$ is a σ -algebra.

Proposition 2.1.1. *Let $\mathcal{A}$ be an algebra of sets.*

- (1) $\mathcal{A}$ is closed under finite intersection.
- (2) $\mathcal{A}$ is closed under set difference.
- (3) $\emptyset \in \mathcal{A}$ and $X \in \mathcal{A}$.

Proof. (1) Let $E_1, E_2, \dots, E_n \in \mathcal{A}$. Then $\bigcap_{i=1}^n E_i = \left(\bigcup_{i=1}^n E_i^c\right)^c \in \mathcal{A}$. (2) If $E_1, E_2 \in \mathcal{A}$, then $E_1 \setminus E_2 = E_1 \cap E_2^c \in \mathcal{A}$. (3) For any $E \in \mathcal{A}$, we have $\emptyset = E \cap E^c \in \mathcal{A}$ and $X = E \cup E^c \in \mathcal{A}$. $\square$

Proposition 2.1.2. *Let $\mathcal{A}$ be an algebra of sets. The following are equivalent:*

- (1) $\mathcal{A}$ is a σ -algebra.
- (2) $\mathcal{A}$ is closed under countable disjoint union.

Proof. If $\mathcal{A}$ is a σ -algebra, then it is closed under countable union, hence it must be closed under countable disjoint union. Conversely, suppose $\mathcal{A}$ is closed under countable disjoint union. Let $\{E_n\}_{n=1}^\infty$ be a family of sets in $\mathcal{A}$. Define:

$$\begin{aligned} F_1 &= E_1, \\ F_2 &= E_2 \setminus E_1 \\ F_3 &= E_3 \setminus (E_1 \cup E_2) \\ &\vdots \\ F_n &= E_n \setminus (E_1 \cup E_2 \cup \dots \cup E_{n-1}). \end{aligned}$$

Let $i < j$ be natural numbers and consider $F_i \cap F_j$. If $x \in F_i$, then $x \in E_i$. Since $i < j$, we have $x \in E_1 \cup \dots \cup E_{j-1}$, hence $x \notin F_j$. Thus $F_i \cap F_j = \emptyset$. Inductively, $\{F_n\}_{n=1}^\infty$ is a countable family of disjoint sets.

Clearly $\bigcup_{n=1}^\infty F_n \subset \bigcup_{n=1}^\infty E_n$. Conversely, let $x \in \bigcup_{n=1}^\infty E_n$. Find the smallest natural number j such that $x \in E_j$. Then $x \notin E_i$ for $i < j$. Hence $x \in E_j \setminus (E_1 \cup \dots \cup E_{j-1}) = F_j \subset \bigcup_{n=1}^\infty F_n$.

Since $\mathcal{A}$ is closed under countable disjoint unions, we have $\bigcup_{n=1}^\infty E_n = \bigcup_{n=1}^\infty F_n \in \mathcal{A}$. Since $\{E_n\}_{n=1}^\infty$ was an arbitrary family of sets, this means $\mathcal{A}$ is a σ -algebra. $\square$

Proposition 2.1.3. *If $\{\mathcal{E}_i\}_{i \in I}$ is a family of σ -algebras, then $\bigcap_{i \in I} \mathcal{E}_i$ is a σ -algebra.*

Proof. Let $\{E_n\}_{n=1}^\infty$ be a family of sets contained in $\bigcap_{i \in I} \mathcal{E}_i$. Then each $\{E_n\}_{n=1}^\infty$ is contained in $\mathcal{E}_i$. Since $\mathcal{E}_i$ is a σ -algebra for every $i \in I$, we have $\bigcup_{n=1}^\infty E_n \in \mathcal{E}_i$ for every $i \in I$, which means $\bigcup_{n=1}^\infty E_n \in \bigcap_{i \in I} \mathcal{E}_i$.

Similarly, let $F \in \bigcap_{i \in I} \mathcal{E}_i$. Then $F \in \mathcal{E}_i$ for every $i \in I$. Since $\mathcal{E}_i$ is a σ -algebra for every $i \in I$, we have $F^c \in \mathcal{E}_i$ for every $i \in I$, which means $F^c \in \bigcap_{i \in I} \mathcal{E}_i$. Thus $\bigcap_{i \in I} \mathcal{E}_i$ is a σ -algebra. $\square$

Definition 2.1.2. Let X be a set and $\mathcal{E} \subset \mathcal{P}(X)$. The σ -algebra generated by $\mathcal{E}$ is $\mathfrak{M}(\mathcal{E}) := \bigcap \{\mathcal{A} \mid \mathcal{E} \subset \mathcal{A}, \mathcal{A} \text{ is a } \sigma\text{-algebra}\}$.

Lemma 2.1.4. *If $\mathcal{E} \subset \mathfrak{M}(\mathcal{F})$, then $\mathfrak{M}(\mathcal{E}) \subset \mathfrak{M}(\mathcal{F})$.*

Proof. Since $\mathfrak{M}(\mathcal{F})$ is a σ -algebra containing $\mathcal{E}$, it must contain $\mathfrak{M}(\mathcal{E})$, as $\mathfrak{M}(\mathcal{E})$ is the smallest σ -algebra generated by $\mathcal{E}$. $\square$

Given a metric space X , we might only be interested in what is generated by its open sets.

Definition 2.1.3. Let X be a metric space and τ_X the set containing all of its open sets. The *Borel σ -algebra on X* is the set $\mathcal{B}_X := \mathfrak{M}(\tau_X)$. Members of $\mathcal{B}_X$ are called *Borel sets*.

Talk about G delta and F sigma sets.

Proposition 2.1.5. $\mathcal{B}_{\mathbb{R}}$ is generated by each of the following:

- (1) the open intervals: $\mathcal{E}_1 = \{(a, b) \mid a < b\}$,
- (2) the closed intervals: $\mathcal{E}_2 = \{[a, b] \mid a < b\}$,
- (3) the half-open intervals: $\mathcal{E}_3 = \{(a, b] \mid a < b\}$ or $\mathcal{E}_4 = \{[a, b) \mid a < b\}$,
- (4) the open rays: $\mathcal{E}_5 = \{(a, \infty) \mid a \in \mathbb{R}\}$ or $\mathcal{E}_6 = \{(-\infty, a) \mid a \in \mathbb{R}\}$,
- (5) the closed rays: $\mathcal{E}_7 = \{(a, \infty) \mid a \in \mathbb{R}\}$ or $\mathcal{E}_8 = \{(-\infty, a) \mid a \in \mathbb{R}\}$.

Proof. We are required to show for each $\mathcal{E}_i$ that $\mathfrak{M}(\mathcal{E}_i) = \mathcal{B}_{\mathbb{R}}$.

(1) Let $I \in \mathcal{E}_1$ be an open interval. Then $I \in \mathcal{B}_{\mathbb{R}}$ because I is an open subset of $\mathbb{R}$. From this we have $\mathcal{E}_1 \subset \mathcal{B}_{\mathbb{R}}$, but because $\mathcal{B}_{\mathbb{R}}$ is a σ -algebra containing $\mathcal{E}_1$, and $\mathfrak{M}(\mathcal{E}_1)$ is the smallest σ -algebra containing $\mathcal{E}_1$, we have $\mathfrak{M}(\mathcal{E}_1) \subset \mathcal{B}_{\mathbb{R}}$. Conversely, let $U \in \tau_{\mathbb{R}}$. Then U is at most the countable union of disjoint open intervals, hence $U \in \mathfrak{M}(\mathcal{E}_1)$. From this we have $\tau_{\mathbb{R}} \subset \mathfrak{M}(\mathcal{E}_1)$, but because $\mathfrak{M}(\mathcal{E}_1)$ is a σ -algebra containing $\tau_{\mathbb{R}}$, and $\mathfrak{M}(\tau_{\mathbb{R}})$ is the smallest σ -algebra containing $\tau_{\mathbb{R}}$, we have $\mathfrak{M}(\tau_{\mathbb{R}}) = \mathcal{B}_{\mathbb{R}} \subset \mathfrak{M}(\mathcal{E}_1)$.

(2) Let $I \in \mathcal{E}_1$ be an open interval. Then I^c is a closed interval, hence it is contained in $\mathfrak{M}(\mathcal{E}_2)$. Whence $\mathfrak{M}(\mathcal{E}_1) \subset \mathfrak{M}(\mathcal{E}_2)$. Conversely, let $C \in \mathcal{E}_2$ be a closed interval. Then C can be written as the countable intersection of open sets. This means $C \in \mathfrak{M}(\mathcal{E}_1)$. Whence $\mathfrak{M}(\mathcal{E}_2) \subset \mathfrak{M}(\mathcal{E}_1)$. Thus $\mathfrak{M}(\mathcal{E}_2) = \mathfrak{M}(\mathcal{E}_1) = \mathcal{B}_{\mathbb{R}}$ $\square$

Given two sets X_1 and X_2 with respective σ -algebras $\mathcal{M}_1$ and $\mathcal{M}_2$, we'd like to better understand what a σ -algebra on $X_1 \times X_2$ looks like.

Definition 2.1.4. Let $\{X_\alpha\}_{\alpha \in I}$ be an indexed family of nonempty sets, $X = \prod_{\alpha \in I} X_\alpha$, and $\pi_\alpha : X \rightarrow X_\alpha$ the projection maps. If $\mathcal{M}_\alpha$ is a σ -algebra on X_α for each α , the *product σ -algebra* on X is:

$$\bigotimes_{\alpha \in I} \mathcal{M}_\alpha := \mathfrak{M}\left(\{\pi_\alpha^{-1}(E_\alpha) \mid E_\alpha \in \mathcal{M}_\alpha, \alpha \in I\}\right)$$

The significance of this definition will become clear later. If X_1 and X_2 are sets with σ -algebras $\mathcal{M}_1$ and $\mathcal{M}_2$ respectively, one might wonder a more intuitive definition for the σ -algebra on $X_1 \times X_2$ to be:

$$\mathfrak{M}\left(\{E_1 \times E_2 \mid E_1 \in \mathcal{M}_1, E_2 \in \mathcal{M}_2\}\right).$$

In fact, these two definitions are equivalent if we're only taking the cartesian product of countably many sets! Before proving this, note that if $E_i \subset X_i$ and $\pi_i : X_1 \times X_2 \rightarrow X_i$ for $i = 1, 2$, then:

$$\begin{aligned} \pi_1^{-1}(E_1) &= E_1 \times X_2, \\ \pi_2^{-1}(E_2) &= X_1 \times E_2. \end{aligned}$$

We can see that:

$$\begin{aligned}
 \pi_1^{-1}(E_1) \cap \pi_2^{-1}(E_2) &= (E_1 \times X_2) \cap (X_1 \times E_2) \\
 &= \{(e_1, x_2) \mid e_1 \in E_1, x_2 \in X_2\} \cap \{(x_1, e_2) \mid x_1 \in X_1, e_2 \in E_2\} \\
 &= \{(x, y) \mid (x \in E_1 \wedge x \in X_1) \wedge (y \in X_2 \wedge y \in E_2)\} \\
 &= \{(x, y) \mid x \in E_1 \wedge y \in E_2\} \\
 &= E_1 \times E_2.
 \end{aligned}$$

Proposition 2.1.6. *Let A be a countable indexing set, $\{X_\alpha\}_{\alpha \in A}$ an indexed family of nonempty sets, $X = \prod_{\alpha \in A} X_\alpha$, $\pi_\alpha : X \rightarrow X_\alpha$ the projection maps, and $\mathcal{M}_\alpha$ a σ -algebra on X_α for each $\alpha \in A$. Then $\bigotimes_{\alpha \in A} \mathcal{M}_\alpha$ is generated by:*

$$\left\{ \prod_{\alpha \in A} E_\alpha \mid E_\alpha \in \mathcal{M}_\alpha \right\}$$

Proof. Denote $\mathcal{O}$ as the σ -algebra generated by $\{\prod_{\alpha \in A} E_\alpha \mid E_\alpha \in \mathcal{M}_\alpha\}$. If $E_\alpha \in \mathcal{M}_\alpha$, then $\pi_\alpha^{-1}(E_\alpha) = \prod_{\beta \in A} E_\beta$, where $E_\beta = X_\beta$ for $\beta \neq \alpha$. This gives the inclusion:

$$\{\pi_\alpha^{-1}(E_\alpha) \mid E_\alpha \in \mathcal{M}_\alpha, \alpha \in I\} \subset \mathcal{O}.$$

But since $\bigotimes_{\alpha \in A} \mathcal{M}_\alpha$ is the smallest σ -algebra generated by $\{\pi_\alpha^{-1}(E_\alpha) \mid E_\alpha \in \mathcal{M}_\alpha, \alpha \in I\}$, we get:

$$\bigotimes_{\alpha \in A} \mathcal{M}_\alpha \subset \mathcal{O}.$$

Conversely, note that $\prod_{\alpha \in A} E_\alpha = \bigcap_{\alpha \in A} \pi_\alpha^{-1}(E_\alpha)$. Since each $\pi_\alpha^{-1}(E_\alpha)$ is in $\bigotimes_{\alpha \in A} \mathcal{M}_\alpha$, and since $\bigotimes_{\alpha \in A} \mathcal{M}_\alpha$ is closed under countable intersections, we have that $\prod_{\alpha \in A} E_\alpha \in \bigotimes_{\alpha \in A} \mathcal{M}_\alpha$. Hence we have the following inclusion:

$$\left\{ \prod_{\alpha \in A} E_\alpha \mid E_\alpha \in \mathcal{M}_\alpha \right\} \subset \bigotimes_{\alpha \in A} \mathcal{M}_\alpha.$$

But since $\mathcal{O}$ is the smallest σ -algebra generated by $\{\prod_{\alpha \in A} E_\alpha \mid E_\alpha \in \mathcal{M}_\alpha\}$, it must be the case that $\mathcal{O} \subset \bigotimes_{\alpha \in A} \mathcal{M}_\alpha$. Since we showed inclusion in both directions, $\mathcal{O} = \bigotimes_{\alpha \in A} \mathcal{M}_\alpha$; i.e., $\bigotimes_{\alpha \in A} \mathcal{M}_\alpha$ is generated by $\{\prod_{\alpha \in A} E_\alpha \mid E_\alpha \in \mathcal{M}_\alpha\}$. $\square$

If there exists a set $\mathcal{E}_\alpha$ such that $\mathcal{M}_\alpha = \mathfrak{M}(\mathcal{E}_\alpha)$, then we can generate the product similar algebra in a different way.

Lemma 2.1.7. **** Let $\{X_\alpha\}_{\alpha \in A}$ be a family of nonempty sets, $\mathcal{E}_\alpha \subset \mathcal{P}(X_\alpha)$, and $\mathcal{M}_\alpha := \mathfrak{M}(\mathcal{E}_\alpha)$ for each $\alpha \in A$. Then $\bigotimes_{\alpha \in A} \mathcal{M}_\alpha$ is generated by:*

$$\mathcal{F}_1 = \{\pi_\alpha^{-1}(E_\alpha) \mid E_\alpha \in \mathcal{E}_\alpha, \alpha \in A\}.$$

If A is countable and $X_\alpha \in \mathcal{E}_\alpha$ for each α , then $\bigotimes_{\alpha \in A} \mathcal{M}_\alpha$ is generated by:

$$\mathcal{F}_2 = \left\{ \prod_{\alpha \in A} E_\alpha \mid E_\alpha \in \mathcal{E}_\alpha \right\}.$$

Proof. Automatically we have $\mathcal{F}_1 \subset \bigotimes_{\alpha \in A} \mathcal{M}$, hence $\mathfrak{M}(\mathcal{F}_1) \subset \bigotimes_{\alpha \in A} \mathcal{M}_\alpha$. Conversely, consider the set $S := \{E \subset X_\alpha \mid \pi_\alpha^{-1}(E) \in \mathfrak{M}(\mathcal{F}_1)\}$. This set is a σ -algebra. Indeed, if $\{E_n\}_{n \in \mathbb{N}}$ is a family of elements of S , then $\{\pi_\alpha^{-1}(E_n)\}_{n \in \mathbb{N}}$ is a family of elements of $\mathfrak{M}(\mathcal{F}_1)$. Since $\mathfrak{M}(\mathcal{F}_1)$ is a σ -algebra, we get:

$$\bigcup_{n \in \mathbb{N}} \pi_\alpha^{-1}(E_n) = \pi_\alpha^{-1}\left(\bigcup_{n \in \mathbb{N}} E_n\right).$$

Hence $\bigcup_{n \in \mathbb{N}} E_n \in S$. A similar argument can be used to show S is closed under complements. With this in mind, note that we also have $\mathcal{E}_\alpha \subset \mathfrak{M}(\mathcal{F}_1)$. Indeed, for any $E \in \mathcal{E}_\alpha$,
would like to finish some day... $\square$

This result helps us prove an interesting fact about Borel sets.

Corollary 2.1.8. *** Let $X_1, \dots, X_n$ be metric spaces and let $X = \prod_{j=1}^n X_j$, equipped with the product metric. Then $\bigotimes_{j=1}^n \mathcal{B}_{X_j} \subset \mathcal{B}_X$. If the X_j 's are separable, then $\bigotimes_{j=1}^n \mathcal{B}_{X_j} = \mathcal{B}_X$.

Proof. would like to finish some day... $\square$

Corollary 2.1.9. *** $\mathcal{B}_{\mathbb{R}^n} = \bigotimes_{j=1}^n \mathcal{B}_{\mathbb{R}}$.

Proof. would like to finish some day... $\square$

§ 2.2. Measures

Definition 2.2.1. Let X be a nonempty set and $\mathcal{M}$ a σ -algebra on X . A *measure* is a function:

$$\mu : \mathcal{M} \rightarrow [0, \infty]$$

satisfying:

$$(1) \mu(\emptyset) = 0;$$

$$(2) \text{ if } \{E_j\}_{j=1}^\infty \subset \mathcal{M} \text{ are pairwise disjoint, then } \mu\left(\bigcup_{j=1}^\infty E_j\right) = \sum_{j=1}^\infty \mu(E_j).$$

We call the ordered triple $(X, \mathcal{M}, \mu)$ a *measure space*. The sets $E \in \mathcal{M}$ are called *measurable sets*. If $\mu(X) < \infty$, then μ is called *finite*. If $X = \bigcup_{j=1}^\infty E_j$, $E_j \in \mathcal{M}$ for all j , and $\mu(E_j) < \infty$, then μ is *σ -finite*.

Example 2.2.1.

(1) Let X be a nonempty set. Define $\mu : \mathcal{P}(X) \rightarrow [0, \infty]$ by $\mu(E) = \sum_{x \in E} 1$ (note that this series exists as our codomain is the extended real numbers). Then $(X, \mathcal{P}(X), \mu)$ is a measure space, and we call μ the *counting measure*.

(2) Let X be nonempty and fix a point $x_0 \in X$. Define $\mu : \mathcal{P}(X) \rightarrow [0, \infty]$ by:

$$\mu(E) = \begin{cases} 0, & x_0 \notin E \\ 1, & x_0 \in E. \end{cases}$$

Then $(X, \mathcal{M}, \mu)$ is a measure space, and we call μ the *dirac measure*.

- (3) Examples (1) and (2) can be generalized. Let X be a nonempty set and $f : X \rightarrow [0, \infty]$. Define $\mu : \mathcal{P}(X) \rightarrow [0, \infty]$ by $\mu(E) = \sum_{x \in E} f(x)$. Then $(X, \mathcal{P}(X), \mu)$ is a measure space. If $f : X \rightarrow [0, \infty]$ is the identity function, then μ is the counting measure. If $f : X \rightarrow [0, \infty]$ is the indicator function $\mathbb{1}_{\{x_0\}}$, then μ is the dirac measure.
- (4) Let X be an uncountable set and let $\mathcal{M}$ be the σ -algebra of countable or co-countable sets. Define $\mu : \mathcal{M} \rightarrow [0, \infty]$ by:

$$\mu(E) = \begin{cases} 0, & E \text{ is countable} \\ 1, & E \text{ is co-countable.} \end{cases}$$

Then $(X, \mathcal{M}, \mu)$ is a measure space.

Proposition 2.2.1. *Let $(X, \mathcal{M}, \mu)$ be a measure space and $E, F \in \mathcal{M}$. Then $\mu(E) + \mu(F) = \mu(E \cup F) + \mu(E \cap F)$.*

Proof. (1) We can write $E \cup F = (E \setminus F) \sqcup (E \cap F) \sqcup (F \setminus E)$. This means $\mu(E \cup F) = \mu(E \setminus F) + \mu(E \cap F) + \mu(F \setminus E)$. This is equivalent to:

$$\begin{aligned} \mu(E \cup F) &= \left(\mu(E \setminus F) + \mu(E \cap F) \right) + \left(\mu(F \setminus E) + \mu(E \cap F) \right) - \mu(E \cap F) \\ &= \mu\left((E \setminus F) \cup (E \cap F)\right) + \mu\left((F \setminus E) \cup (E \cap F)\right) - \mu(E \cap F) \\ &= \mu(E) + \mu(F) - \mu(E \cap F) \end{aligned}$$

Hence $\mu(E) + \mu(F) = \mu(E \cup F) + \mu(E \cap F)$. □

Proposition 2.2.2. *Let $(X, \mathcal{M}, \mu)$ be a measure space.*

- (1) *If $E, F \in \mathcal{M}$ and $E \subset F$, then $\mu(E) \leq \mu(F)$ (Monotonicity).*
- (2) *Let $\{E_i\}_{i=1}^{\infty} \subset \mathcal{M}$. Then $\mu\left(\bigcup_{i=1}^{\infty} E_i\right) \leq \sum_{i=1}^{\infty} \mu(E_i)$ (Subadditivity).*
- (3) *Let $\{E_i\}_{i=1}^{\infty} \subset \mathcal{M}$ with $E_1 \subset E_2 \subset E_3 \subset \dots$. Then $\mu\left(\bigcup_{i=1}^{\infty} E_i\right) = \lim_{i \rightarrow \infty} \mu(E_i)$ (Continuity from below).*
- (4) *Let $\{E_i\}_{i=1}^{\infty} \subset \mathcal{M}$ with $E_1 \supset E_2 \supset E_3 \supset \dots$ provided $\mu(E_1) < \infty$. Then $\mu\left(\bigcap_{i=1}^{\infty} E_i\right) = \lim_{i \rightarrow \infty} \mu(E_i)$.*

Proof. (1) We can write $F = E \sqcup (F \setminus E)$, giving $\mu(F) = \mu(E \sqcup (F \setminus E)) = \mu(E) + \mu(F \setminus E) \geq \mu(E)$.

(2) Given $\{E_i\}_{i=1}^{\infty} \subset \mathcal{M}$, define $F_i = E_i \setminus \bigcup_{k=1}^{i-1} E_k$ for each i . Then $\{F_i\}_{i=1}^{\infty}$ is a sequence of pairwise disjoint subsets of X . In particular, we have $F_i \subset E_i$ for each i , and $\bigcup_{i=1}^{\infty} E_i = \bigcup_{i=1}^{\infty} F_i$. Together:

$$\begin{aligned} \mu\left(\bigcup_{i=1}^{\infty} E_i\right) &= \mu\left(\bigcup_{i=1}^{\infty} F_i\right) \\ &= \sum_{i=1}^{\infty} \mu(F_i) \\ &\leq \sum_{i=1}^{\infty} \mu(E_i). \end{aligned}$$

(3) Define $E_0 = \emptyset$. Then $\{E_i \setminus E_{i-1}\}_{i=1}^{\infty}$ will be a sequence of pairwise disjoint subsets of X with the property that $\bigcup_{i=1}^{\infty} E_i = \bigcup_{i=1}^{\infty} E_i \setminus E_{i-1}$. With this:

$$\begin{aligned} \mu\left(\bigcup_{i=1}^{\infty} E_i\right) &= \mu\left(\bigcup_{i=1}^{\infty} E_i \setminus E_{i-1}\right) \\ &= \sum_{i=1}^{\infty} \mu(E_i \setminus E_{i-1}). \end{aligned}$$

Note that $E_i = E_{i-1} \sqcup (E_i \setminus E_{i-1})$. This means $\mu(E_i) = \mu(E_{i-1}) + \mu(E_i \setminus E_{i-1})$, which is equivalent to $\mu(E_i \setminus E_{i-1}) = \mu(E_i) - \mu(E_{i-1})$. Hence:

$$\begin{aligned} \mu\left(\bigcup_{i=1}^{\infty} E_i\right) &= \sum_{i=1}^{\infty} \mu(E_i \setminus E_{i-1}) \\ &= \lim_{i \rightarrow \infty} \sum_{n=1}^i \mu(E_i \setminus E_{i-1}) \\ &= \lim_{i \rightarrow \infty} \sum_{n=1}^i \mu(E_i) - \mu(E_{i-1}) \\ &= \lim_{i \rightarrow \infty} \mu(E_i). \end{aligned}$$

(4) Define $F_i = E_1 \setminus E_i$. Then $\{F_i\}_{i=1}^{\infty}$ will be a sequence of increasing unions of subsets of X . In particular, we have that $\bigcup_{i=1}^{\infty} F_i = E_1 \setminus \bigcap_{i=1}^{\infty} E_i$. This means:

$$\begin{aligned} E_1 &= \left(E_1 \setminus \bigcap_{i=1}^{\infty} E_i\right) \sqcup \left(\bigcap_{i=1}^{\infty} E_i\right) \\ &= \left(\bigcup_{i=1}^{\infty} F_i\right) \sqcup \left(\bigcap_{i=1}^{\infty} E_i\right). \end{aligned}$$

Hence by (3):

$$\begin{aligned} \mu\left(\bigcap_{i=1}^{\infty} E_i\right) &= \mu(E_1) - \mu\left(\bigcup_{i=1}^{\infty} F_i\right) \\ &= \mu(E_1) - \lim_{i \rightarrow \infty} \mu(F_i). \end{aligned}$$

But $F_i = E_1 \setminus E_i$, so $E_1 = F_i \sqcup E_i$. This gives $\mu(E_1) = \mu(F_i) + \mu(E_i)$, hence:

$$\begin{aligned} \mu\left(\bigcap_{i=1}^{\infty} E_i\right) &= \mu(E_1) - \lim_{i \rightarrow \infty} \mu(F_i) \\ &= \mu(E_1) - \lim_{i \rightarrow \infty} (\mu(E_1) - \mu(E_i)) \\ &= \lim_{i \rightarrow \infty} \mu(E_i). \end{aligned}$$

□

Definition 2.2.2. Let $(X, \mathcal{M}, \mu)$ be a measure space.

- (1) If $E \in \mathcal{M}$ is a measurable set such that $\mu(E) = 0$, then E is called a *null-set*.
- (2) If a property holds on all of X except on a null-set, then it holds *almost everywhere* (abbreviated *a.e.*).
- (3) If $\mathcal{M}$ has the property that every subset of a null-set is measurable, then $\mathcal{M}$ is said to be *complete*.

Theorem 2.2.3. *Let $(X, \mathcal{M}, \mu)$ be a measure space. Let $\mathcal{N} := \{E \in \mathcal{M} \mid \mu(E) = 0\}$. Then the set $\overline{\mathcal{M}} := \{E \cup F \mid E \in \mathcal{M}, F \subset N \text{ for some } N \in \mathcal{N}\}$ is a σ -algebra. Moreover, there exists a unique function $\overline{\mu} : \overline{\mathcal{M}} \rightarrow [0, \infty]$ which extends μ .*

Proof. Consider the family $\{E_i \cup F_i\}_{i=1}^{\infty} \subset \overline{\mathcal{M}}$, where for each i , there exists $N_i \in \mathcal{N}$ such that $F_i \subset N_i$. Then:

$$\bigcup_{i=1}^{\infty} E_i \cup F_i = \left(\bigcup_{i=1}^{\infty} E_i \right) \cup \left(\bigcup_{i=1}^{\infty} F_i \right);$$

Since $\mathcal{M}$ is a σ -algebra, we have $\bigcup_{i=1}^{\infty} E_i \in \mathcal{M}$. Moreover, $\bigcup_{i=1}^{\infty} F_i \subset \bigcup_{i=1}^{\infty} N_i$. By subadditivity, we have $\mu\left(\bigcup_{i=1}^{\infty} N_i\right) \leq \sum_{i=1}^{\infty} \mu(N_i) = 0$. Hence $\bigcup_{i=1}^{\infty} N_i \in \mathcal{N}$. Together, this shows that $\bigcup_{i=1}^{\infty} E_i \cup F_i \in \overline{\mathcal{M}}$.

Now let $E \cup F \in \overline{\mathcal{M}}$. Then:

$$\begin{aligned} (E \cup F)^c &= \left((E \cup N) \cap (N^c \cup F) \right)^c \\ &= (E \cup N)^c \cup (N^c \cup F)^c \\ &= (E^c \cap N^c) \cup (N \cap F^c) \\ &= (E^c \cap N^c) \cup (N \setminus F). \end{aligned}$$

Since $E^c, N^c \in \mathcal{M}$, and since $\mathcal{M}$ is closed under countable intersections, we have $E^c \cap N^c \in \mathcal{M}$. Since $N \in \mathcal{N}$, surely $N \setminus F \in \mathcal{N}$, whence $(E \cup F)^c \in \overline{\mathcal{M}}$. Thus $\overline{\mathcal{M}}$ is a σ -algebra.

Define $\overline{\mu} : \overline{\mathcal{M}} \rightarrow [0, \infty]$ by $\overline{\mu}(E \cup F) = \mu(E)$. We must first show that this map is well-defined. Suppose $E_1 \cup F_1 = E_2 \cup F_2$, where $F_i \subset N_i \in \mathcal{N}$ for $i = 1, 2$. Then $E_1 \cup F_1 \subset E_2 \cup N_2$, hence $E_1 \subset E_2 \cup N_2$. This means $\mu(E_1) \leq \mu(E_2 \cup N_2) \leq \mu(E_2) + \mu(N_2) = \mu(E_2)$. But an identical argument gives $E_2 \subset E_1 \cup N_1$, hence $\mu(E_2) \leq \mu(E_1 \cup N_1) \leq \mu(E_1) + \mu(N_1) = \mu(E_1)$. Together, $\mu(E_1) = \mu(E_2)$, allowing us to show that:

$$\begin{aligned} \overline{\mu}(E_1 \cup F_1) &= \mu(E_1) \\ &= \mu(E_2) \\ &= \overline{\mu}(E_2 \cup F_2). \end{aligned}$$

Thus $\overline{\mu}$ is well-defined. Note that $\emptyset \in \mathcal{M}$ and $\emptyset \in \mathcal{N}$, hence $\overline{\mu}(\emptyset) = \mu(\emptyset) = 0$. Now if $\{E_i \cup F_i\}_{i=1}^{\infty} \subset \overline{\mathcal{M}}$ are pairwise disjoint, then it must be the case that $\{E_i\}_{i=1}^{\infty}$ are pairwise disjoint. This gives:

$$\overline{\mu} \left(\bigcup_{i=1}^{\infty} E_i \cup F_i \right) = \overline{\mu} \left(\bigcup_{i=1}^{\infty} E_i \cup \bigcup_{i=1}^{\infty} F_i \right).$$

Note that $\bigcup_{i=1}^{\infty} E_i \in \mathcal{M}$, and $\bigcup_{i=1}^{\infty} F_i \subset \bigcup_{i=1}^{\infty} N_i \in \mathcal{N}$, hence:

$$\begin{aligned} \bar{\mu}\left(\bigcup_{i=1}^{\infty} E_i \cup F_i\right) &= \bar{\mu}\left(\bigcup_{i=1}^{\infty} E_i \cup \bigcup_{i=1}^{\infty} F_i\right) \\ &= \mu\left(\bigcup_{i=1}^{\infty} E_i\right) \\ &\leq \sum_{i=1}^{\infty} \mu(E_i) \\ &= \sum_{i=1}^{\infty} \bar{\mu}(E_i \cup F_i). \end{aligned}$$

Thus $\bar{\mu}$ is a measure. Moreover, it is an extension of μ since for any $E \in \mathcal{M}$, $\bar{\mu}(E) = \bar{\mu}(E \cup \emptyset) = \mu(E)$. It remains to show that this extension is unique. Suppose γ is also an extension of μ onto $\bar{\mathcal{M}}$. (WTS $\gamma(\bar{\mathcal{M}}) = \bar{\mu}(\bar{\mathcal{M}})$). Let $E \cup F \in \bar{\mathcal{M}}$, where $F \subset N \in \mathcal{N}$. By subadditivity:

$$\begin{aligned} \gamma(E \cup F) &\leq \gamma(E) \cup \gamma(F) \\ &= \gamma(E) \\ &= \mu(E) \quad \text{Since } \gamma|_{\mathcal{M}} = \mu \\ &= \bar{\mu}(E \cup F). \end{aligned}$$

But by monotonicity:

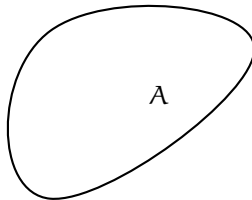
$$\begin{aligned} \bar{\mu}(E \cup F) &= \mu(E) \\ &= \gamma(E) \quad \text{Since } \gamma|_{\mathcal{M}} = \mu \\ &\leq \gamma(E \cup F). \end{aligned}$$

Hence $\gamma(E \cup F) = \bar{\mu}(E \cup F)$. Since γ and $\bar{\mu}$ are equal on all values of $\bar{\mathcal{M}}$, they are equal. Thus $\bar{\mu}$ is unique. $\square$

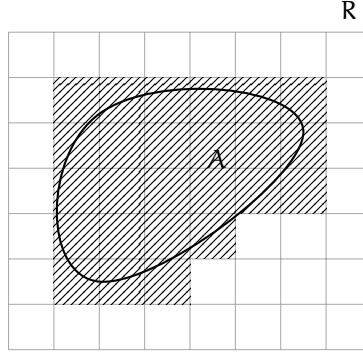
Definition 2.2.3. Let $(X, \mathcal{M}, \mu)$ be a measure space. The measure $\bar{\mu}$ is called the *completion* of μ , and $\bar{\mathcal{M}}$ is called the *completion of $\mathcal{M}$ with respect to μ* .

§ 2.3. Outer Measures

Consider the set $A \subset \mathbb{R}^2$ below:



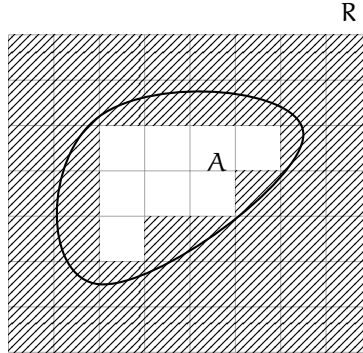
What would the "area" for this set be? Rectangles are very easy to compute, so let's cover A with an evenly partitioned rectangle R :



We can define the "outer area" of our set A as:

$$\text{outerarea}(A) = \left\{ \sum R_j \mid R_j \subset R \text{ is a rectangle covering } A \right\}.$$

Now consider the outer area of the set $R \setminus A$:



We can define the "inner area" of A as:

$$\text{innerarea}(A) = \text{area}(R) - \text{outerarea}(R \setminus A)$$

Taking finer and finer partitions, if the inner area of A is equal to the outer area of A , we can define them as the *area* of A . We can abstract this notion of outer area as follows.

Definition 2.3.1. Let X be a nonempty set. An *outer measure* on X is a function:

$$\mu^* : \mathcal{P}(X) \rightarrow [0, \infty]$$

satisfying:

- (1) $\mu^*(\emptyset) = 0$;
- (2) If $A \subset B$, then $\mu^*(A) \leq \mu^*(B)$ (Monotonicity);
- (3) $\mu^*\left(\bigcup_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty} \mu^*(A_i)$ (Subadditivity).

(1) is clear otherwise we reach a contradiction. (2) follows from the fact that if we have rectangles covering B , then surely the area of those rectangles must be greater than or equal to the area of the rectangles covering A . (3) comes from the fact that if we have two overlapping sets, there may be a rectangle which covers both sets simultaneously, hence adding up the rectangles of their union is "more precise" than adding up the rectangles covering each set individually. We will prove rigorously that our motivating example does in fact give us an outer measure.

Proposition 2.3.1. *Let X be any nonempty set and $\mathcal{E} \subset \mathcal{P}(X)$, with $\emptyset, X \in \mathcal{E}$. Let $\rho : \mathcal{E} \rightarrow [0, \infty]$ be a function such that $\rho(\emptyset) = 0$. The function $\mu^* : \mathcal{P}(X) \rightarrow [0, \infty]$ defined by:*

$$\mu^*(A) = \inf \left\{ \sum_{i=1}^{\infty} \rho(E_i) \mid \bigcup_{i=1}^{\infty} E_i \supset A, E_i \in \mathcal{E} \right\}$$

is an outer measure on X .

Proof. Since $\emptyset \subset \emptyset$ and $\emptyset \in \mathcal{E}$, we get $\mu^*(\emptyset) = 0$. If $A \subset B$ and $\bigcup_{i=1}^{\infty} E_i \supset B$, then surely $\bigcup_{i=1}^{\infty} E_i \supset A$. This gives the containment:

$$\left\{ \sum_{i=1}^{\infty} \rho(E_i) \mid \bigcup_{i=1}^{\infty} E_i \supset B, E_i \in \mathcal{E} \right\} \supset \left\{ \sum_{i=1}^{\infty} \rho(E_i) \mid \bigcup_{i=1}^{\infty} E_i \supset A, E_i \in \mathcal{E} \right\}.$$

Any lowerbound for the left set will be a lowerbound for the right set, hence $\mu^*(A) \leq \mu^*(B)$. It remains to show that μ^* is subadditive. Let $\{A_j\}_{j=1}^{\infty} \subset \mathcal{P}(X)$. Let $\epsilon > 0$. For each $j = 1, 2, 3, \dots$ find a family $\{E_{i,j}\}_{i=1}^{\infty} \subset \mathcal{E}$ such that:

$$\sum_{i=1}^{\infty} \rho(E_{i,j}) \leq \frac{\epsilon}{2^j} + \mu^*(A_j). \quad \text{Property of infimums.}$$

Note that $A_j \subset \bigcup_{i=1}^{\infty} E_{i,j}$, hence $\{E_{i,j}\}_{i,j \in \mathbb{N}}$ covers $\bigcup_{j=1}^{\infty} A_j$. This means:

$$\begin{aligned} \mu^* \left(\bigcup_{j=1}^{\infty} A_j \right) &\leq \sum_{j=1}^{\infty} \sum_{i=1}^{\infty} \rho(E_{i,j}) \quad \text{Surely any cover will be bigger than the g.l.b.} \\ &\leq \sum_{j=1}^{\infty} \left(\mu^*(A_j) + \frac{\epsilon}{2^j} \right) \quad \text{Also, rearrangements of sums follows from being non-negative} \\ &= \sum_{j=1}^{\infty} \mu^*(A_j) + \epsilon. \end{aligned}$$

Taking $\epsilon \rightarrow \infty$ gives $\mu^* \left(\bigcup_{j=1}^{\infty} A_j \right) \leq \sum_{j=1}^{\infty} \mu^*(A_j)$. □

This next definition will lead us to constructing measures from outer measures.

Definition 2.3.2. Let μ^* be an outer measure on X . A set $A \subset X$ is called μ^* -measurable if, for all $E \subset X$:

$$\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c).$$

This definition is very unintuitive. From our example in $\mathbb{R}^2$, given any rectangle R covering $A \subset \mathbb{R}^2$, we defined the inner area of A as:

$$\text{innerarea}(A) = \text{area}(R) - \text{outerarea}(R \setminus A),$$

where R is a rectangle covering our set A . If the inner area and the outer area are equal, then the above equation becomes:

$$\text{outerarea}(A) = \text{area}(R) - \text{outerarea}(R \setminus A).$$

Rearranging yields:

$$\text{area}(R) = \text{outerarea}(A) + \text{outerarea}(R \setminus A).$$

So under the assumption that the inner and outer area of A is equal, we gain the ability to measure any rectangle R using *only* the parts which intersect A and the parts which intersect A^c . In this case A is a "nice set" that we can use to measure other things.

Lemma 2.3.2 (Caratheodory's Lemma). **** Let μ^* be an outer measure on X . Let $\mathcal{M} = \{A \subset X \mid A \text{ is } \mu^* \text{ measurable}\}$. Then $\mathcal{M}$ is a σ -algebra, and $\mu^*|_{\mathcal{M}}$ is a complete measure (i.e., $(X, \mathcal{M}, \mu^*|_{\mathcal{M}})$ is a complete measure space).*

Proof. 10/2/25

□

Carathéodory's lemma is an extremely powerful result. We will use it to show that a measure defined on an *algebra* can be extended to a measure on a σ -algebra. Typically it's easier to construct algebras than σ -algebras, and a direct application of this will be seen in the next section.

Definition 2.3.3. Let X be a nonempty set. Let $\mathcal{A} \subset \mathcal{P}(X)$ be an algebra of sets. A function $\mu_0 : \mathcal{A} \rightarrow [0, \infty]$ is a *premeasure* if:

- (1) $\mu_0(\emptyset) = 0$;
- (2) If $\{E_i\}_{i=1}^{\infty} \subset \mathcal{A}$ is a family of pairwise disjoint sets with $\bigcup_{i=1}^{\infty} E_i \in \mathcal{A}$, then $\mu_0\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} \mu_0(E_i)$.

Proposition 2.3.3. *If μ_0 is a premeasure on $\mathcal{A}$ and μ^* is the outer measure induced by μ_0 , then $\mu^*|_{\mathcal{A}} = \mu_0$.*

Proof. Let $E \in \mathcal{A}$... **finish some day...**

□

Theorem 2.3.4 (Carathéodory's Extension Theorem). **** Let μ_0 be a premeasure on an algebra $\mathcal{A} \subset \mathcal{P}(X)$. Then:*

- (1) μ_0 can be extended to a measure $\mu : \mathfrak{M}(\mathcal{A}) \rightarrow [0, \infty]$.
- (2) If ν is another measure that extends μ_0 , then $\nu(E) \leq \mu(E)$ for all $E \in \mathfrak{M}(\mathcal{A})$, with equality when $\mu(E) < \infty$.
- (3) If μ_0 is σ -finite, then μ is the unique extension of μ_0 to a measure on $\mathfrak{M}(\mathcal{A})$.

Proof. (1) Our first step will be to build an outer measure on $\mathcal{P}(X)$. We will use the sets from $\mathcal{A}$ as our "measuring sets". For $E \subset X$, define:

$$\mu^*(E) = \inf \left\{ \sum_{j=1}^{\infty} \mu_0(A_j) \mid E \subset \bigcup_{j=1}^{\infty} A_j, \{A_j\}_{j=1}^{\infty} \subset \mathcal{A} \right\}$$

Note that we can trivially cover E with $X \in \mathcal{A}$, hence it is nonempty. By Proposition 2.3.1, the function μ^* is an outer measure. By Caratheodory's Lemma, we have a measure $\mu^*|_{\mathcal{M}^*}$ defined on:

$$\mathcal{M}^* = \{A \subset X \mid A \text{ is } \mu^*\text{-measurable}\}$$

Now that we've constructed $(X, \mathcal{M}^*, \mu^*|_{\mathcal{M}^*})$, our next step is to check that $\mathcal{A} \subset \mathcal{M}^*$ —that is, for any $A \in \mathcal{A}$ and any $E \subset X$, we can write:

$$\mu^*(E) = \mu(E \cap A) + \mu(E \cap A^c).$$

Let $\epsilon > 0$. There exists $\{A_j\}_{j=1}^{\infty} \subset \mathcal{A}$ such that:

$$\begin{aligned} \mu^*(E) + \epsilon &\geq \sum_{j=1}^{\infty} \mu_0(A_j) && \text{Infimum property} \\ &= \sum_{j=1}^{\infty} \mu_0(A_j \cap (A \cup A^c)) \\ &= \sum_{j=1}^{\infty} (\mu_0(A_j \cap A) + \mu_0(A_j \cap A^c)) \\ &= \sum_{j=1}^{\infty} \mu_0(A_j \cap A) + \sum_{j=1}^{\infty} \mu_0(A_j \cap A^c). \end{aligned}$$

Note that $\{A_j \cap A\}_{j=1}^{\infty} \subset \mathcal{A}$ covers $E \cap A$ and $\{A_j \cap A^c\}_{j=1}^{\infty} \subset \mathcal{A}$ covers $E \cap A^c$. Hence:

$$\begin{aligned} \mu^*(E) + \epsilon &\geq \sum_{j=1}^{\infty} \mu_0(A_j \cap A) + \sum_{j=1}^{\infty} \mu_0(A_j \cap A^c) \\ &\geq \mu^*(E \cap A) + \mu^*(E \cap A^c) && \text{Any covering will be larger than g.l.b.} \\ &\geq \mu^*(E). && \text{Monotonicity} \end{aligned}$$

Sending $\epsilon \rightarrow 0$ gives equality, hence A is μ^* -measurable. Having established the inclusion $\mathcal{A} \subset \mathcal{M}^*$, we obtain $\mathfrak{M}(\mathcal{A}) \subset \mathcal{M}^*$.

In summary, we constructed $(X, \mathcal{M}^*, \mu^*|_{\mathcal{M}^*})$ with respect to $\mathcal{A}$, and then showed that $\mathfrak{M}(\mathcal{A}) \subset \mathcal{M}^*$. Claim: The restriction of μ^* to $\mathfrak{M}(\mathcal{A})$ is an extension of μ_0 , that is, for every $A \in \mathcal{A}$ we have $\mu^*|_{\mathfrak{M}(\mathcal{A})}(A) = \mu_0(A)$. For clarity, define $\mu := \mu^*|_{\mathfrak{M}(\mathcal{A})}$.

Let $A \in \mathcal{A}$. Let $\{\mathcal{O}_j\}_{j=1}^{\infty}$ be a family of sets defined as follows: $\mathcal{O}_1 = A$ and $\mathcal{O}_j = \emptyset$ for all $j \neq 1$. Then $\{\mathcal{O}_j\}_{j=1}^{\infty} \subset \mathcal{A}$ is a covering of A , hence $\mu(A) \leq \sum_{j=1}^{\infty} \mu_0(\mathcal{O}_j) = \mu_0(A)$ (again, because $\mu(A)$ is defined as the infimum of $\sum \mu_0(A_j)$ over all covers $\{A_j\}$ of A , every cover

will be greater than or equal to $\mu(A)$). It only remains to show the other direction of the inequality. Let $\epsilon > 0$. Choose a cover $\{A_j\}_{j=1}^\infty \subset \mathcal{A}$ of A . Then by properties of infimums once again, we have:

$$\mu(A) + \epsilon \geq \sum_{j=1}^{\infty} \mu_0(A_j).$$

For each j , define $B_j := A_j \setminus \bigcup_{k=1}^{j-1} A_k \in \mathcal{A}$. This construction was seen in Proposition ??, where we showed that $\{B_j\}_{j=1}^\infty$ is a disjoint family of subsets, $B_j \subset A_j$ for each j , and $\bigcup_{j=1}^\infty B_j = \bigcup_{j=1}^\infty A_j$. Observe that:

$$\begin{aligned} \mu(A) + \epsilon &\geq \sum_{j=1}^{\infty} \mu_0(A_j) \\ &\geq \sum_{j=1}^{\infty} \mu_0(B_j) && \text{Monotonicity of premeasures} \\ &\geq \sum_{j=1}^{\infty} \mu_0(A \cap B_j) && \text{Monotonicity of premeasures} \\ &= \mu_0\left(\bigcup_{j=1}^{\infty} A \cap B_j\right) && \text{Premeasures are countably additive over disjoint sets} \\ &= \mu_0\left(A \cap \bigcup_{j=1}^{\infty} B_j\right) && \text{DeMorgan's Law} \\ &= \mu_0\left(A \cap \bigcup_{j=1}^{\infty} A_j\right) && \text{By Proposition ??} \\ &= \mu_0(A). && \text{Since } \{A_j\}_{j=1}^\infty \text{ is a cover of } A \end{aligned}$$

Taking $\epsilon \rightarrow \infty$ gives $\mu(A) \geq \mu_0(A)$. Thus $\mu(A) = \mu_0(A)$.

(b) Let $E \in \mathfrak{M}(\mathcal{A})$ and $\epsilon > 0$. Choose a cover $\{A_j\}_{j=1}^\infty \subset \mathcal{A}$ of E such that:

$$\begin{aligned} \mu(E) + \epsilon &\geq \sum_{j=1}^{\infty} \mu_0(A_j) \\ &= \sum_{j=1}^{\infty} \mu(A_j). && \text{Since } \mu|_{\mathcal{A}} = \mu_0 \end{aligned}$$

Then:

$$\begin{aligned}
 \nu(E) &\leq \nu\left(\bigcup_{j=1}^{\infty} A_j\right) \\
 &\leq \sum_{j=1}^{\infty} \nu(A_j) \\
 &= \sum_{j=1}^{\infty} \mu_0(A_j) \\
 &\leq \mu(E) + \epsilon.
 \end{aligned}$$

Taking $\epsilon \rightarrow 0$ gives $\nu(E) \leq \mu(E)$. Now suppose $\mu(E) < \infty$. Let $\epsilon > 0$. Find a cover $A := \bigcup_{j=1}^{\infty} A_j \supset E$ such that $\mu(E) + \epsilon \geq \sum_{j=1}^{\infty} \mu_0(A_j) \geq \mu(A)$. Rearranging gives $\mu(A) - \mu(E) \leq \epsilon$, but since $\mu(E) < \infty$ we can write this as $\mu(A \setminus E) \leq \epsilon$. **Finish the rest soon** $\square$

§ 2.4. Measures on $\mathbb{R}$

Carathéodory's Extension Theorem will finally allow us to construct measures on the real number line. As we showed in §2.0, we are unable to define measures on the power set of $\mathbb{R}$. Our game-plan is to find an algebra $\mathcal{A}$ of subsets of $\mathbb{R}$ such that, after applying Carathéodory's Extension Theorem, we obtain a measure on the Borel sets of $\mathbb{R}$.

Definition 2.4.1. Let X be a set. An *elementary family* is a collection $\mathcal{E}$ of subsets of X such that:

- (1) $\emptyset \in \mathcal{E}$;
- (2) if $E, F \in \mathcal{E}$, then $E \cap F \in \mathcal{E}$;
- (3) if $E \in \mathcal{E}$, then E^c is a finite disjoint union of members of $\mathcal{E}$.

Proposition 2.4.1. If $\mathcal{E}$ is an elementary family, the collection $\mathcal{A}$ of finite disjoint unions of members of $\mathcal{E}$ is an algebra.

The proof of this proposition is not relevant to our current discussion.

Definition 2.4.2. An *h-interval* is a left-open, right-closed interval in $\mathbb{R}$ —that is, sets of the form $(a, b]$, (a, ∞) or $\emptyset$.

These h-intervals will be the building blocks for constructing a measure on $\mathbb{R}$. Clearly the intersection of two h-intervals is an h-interval, and the complement of an h-interval is an h-interval or the union of two h-intervals. Moreover, by Proposition 2.4.1, the collection of *finite disjoint unions of h-intervals* is an algebra. This collection can be used to generate $\mathcal{B}_{\mathbb{R}}$, as per Proposition 2.1.5

Proposition 2.4.2. Let μ be a finite measure on $\mathcal{B}_{\mathbb{R}}$. Define $F : \mathbb{R} \rightarrow \mathbb{R}$ by $x \mapsto \mu((-\infty, x])$. Then F is increasing and right-continuous. Moreover, if $a < b$, we have $\mu((a, b]) = F(b) - F(a)$.

Proof. F increasing follows from monotonicity of measures; if $x \leq y$ are real numbers, then it must be that $F(x) = \mu((-\infty, x]) \leq \mu((-\infty, y]) = F(y)$. If $(x_n)_n$ is a sequence in $\mathbb{R}$ such that $(x_n)_n \searrow x$, then:

$$\begin{aligned} F(x) &= \mu((-\infty, x]) \\ &= \mu\left(\bigcap_{n=1}^{\infty} (-\infty, x_n]\right) \\ &= \lim_{n \rightarrow \infty} \mu((-\infty, x_n]) \quad \text{Continuity from above} \\ &= \lim_{n \rightarrow \infty} F(x_n). \end{aligned}$$

Hence F is right-continuous. Finally, if $a < b$, we can write $(-\infty, b] = (-\infty, a] \cup (a, b]$. Since μ is a finite measure, by countable additivity we have $\mu((-\infty, b]) = \mu((-\infty, a]) + \mu((a, b])$. Rearranging gives $\mu((a, b]) = \mu((-\infty, b]) - \mu((-\infty, a]) = F(b) - F(a)$. $\square$

Definition 2.4.3. The function F described in Proposition 2.4.2 is called a *distribution function* of μ .

The distribution function of a measure provides information about the size of a measurable function.

Proposition 2.4.3. Let $\mathcal{A}$ be the collection of finite disjoint unions of h -intervals. Let F be a distribution function of $\mu_0 : \mathcal{A} \rightarrow [0, \infty]$, where μ_0 satisfies the following:

- (1) $\mu_0(\emptyset) = 0$;
- (2) If $\{(a_j, b_j]\}_{j=1}^n$ are disjoint h -intervals, then $\mu_0\left(\bigcup_{j=1}^n (a_j, b_j]\right) = \sum_{j=1}^n (F(b_j) - F(a_j))$.

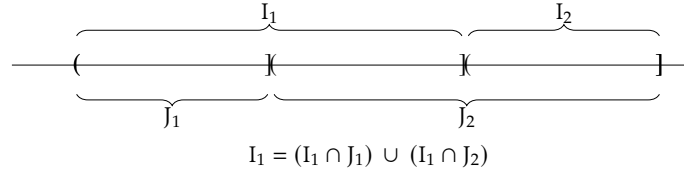
Then μ_0 is a premeasure on the algebra $\mathcal{A}$.

Proof. We must first show that μ_0 is well-defined. We will first prove a special case of μ_0 being well-defined, and then move onto a more general case. Suppose $(a, b] = \bigcup_{j=1}^m (a_j, b_j]$, where each $\{(a_j, b_j]\}_{j=1}^{\infty}$ are disjoint. Without loss of generality, assume $a_1 < a_2 < \dots < a_n$. It must be the case that $b_1 = a_2, b_2 = a_3$, etc. With this, note that the sum $\sum_{j=1}^n (F(b_j) - F(a_j))$ is telescoping, hence $\mu_0\left(\bigcup_{j=1}^n (a_j, b_j]\right) = \sum_{j=1}^n (F(b_j) - F(a_j)) = F(b) - F(a) = \mu_0((a, b])$. Thus, under this special case, μ_0 is well-defined. For the general case, suppose $\{I_i\}_{i=1}^n$ and $\{J_j\}_{j=1}^n$ are collections of disjoint h -intervals such that $\bigcup_{i=1}^n I_i = \bigcup_{j=1}^n J_j$. Note that the intersections $I_i \cap J_j$ are (possibly empty) h -intervals, and they form a disjoint partition of each I_i and each J_j :

$$I_i = \bigcup_{j=1}^n (I_i \cap J_j), \quad J_j = \bigcup_{i=1}^n (I_i \cap J_j).$$

(See Figure 2.1 as a motivating example). Apply the special case inside each I_i and J_j :

$$\mu_0(I_i) = \sum_{j=1}^n \mu_0(I_i \cap J_j), \quad \mu_0(J_j) = \sum_{i=1}^n \mu_0(I_i \cap J_j).$$

Figure 2.1: Illustration of interval intersections between I_1 , I_2 , and J_1 , J_2 .

Summing each equation and rearranging gives $\sum_{i=1}^n \mu_0(I_i) = \sum_{i,j} \mu_0(I_i \cap J_j) = \sum_{j=1}^n \mu_0(J_j)$. Thus μ_0 is well-defined.

We only need to show that μ_0 satisfies countable additivity. Let $\{A_j\}_{j=1}^\infty \subset \mathcal{A}$ be a family of disjoint sets, where $\bigcup_{j=1}^\infty A_j \in \mathcal{A}$. Since everything in $\mathcal{A}$ is a finite disjoint union of h-intervals, we can write:

$$\bigcup_{j=1}^\infty A_j = \bigcup_{n=1}^N I_n, \quad A_j = \bigcup_{i=1}^{N_j} I_{i,j}.$$

But observe that:

$$\begin{aligned} I_n &= I_n \cap \left(\bigcup_{j=1}^\infty A_j \right) \\ &= \bigcup_{j=1}^\infty (I_n \cap A_j) \\ &= \bigcup_{j=1}^\infty \left(I_n \cap \left(\bigcup_{i=1}^{N_j} I_{i,j} \right) \right) \\ &= \bigcup_{j=1}^\infty \bigcup_{i=1}^{N_j} (I_n \cap I_{i,j}). \end{aligned}$$

It is enough to assume that if an h-interval I is a countable disjoint union of h-intervals $\{I_j\}_{j=1}^\infty$, then $\mu_0(I) = \sum_{j=1}^\infty \mu_0(I_j)$. Let $I = (a, b]$ and $I_j = (a_j, b_j]$. One direction of the equality can be seen as follows:

$$\begin{aligned} \mu_0(I) &= \mu_0 \left(\bigcup_{j=1}^n I_j \right) + \mu_0 \left(I \setminus \bigcup_{j=1}^n I_j \right) \\ &\geq \mu_0 \left(\bigcup_{j=1}^n I_j \right) \\ &= \sum_{j=1}^n \mu_0(I_j). \end{aligned}$$

Taking $n \rightarrow \infty$ gives $\mu_0(I) \geq \sum_{j=1}^\infty \mu_0(I_j)$. Conversely, let $\epsilon > 0$. By the right-continuity of F , choose $\delta > 0$ such that $F(a + \delta) - F(a) < \epsilon$. Similarly, choose $\delta_j > 0$ such that

$F(b_j + \delta_j) - F(b_j) < \frac{\epsilon}{2^j}$. Then the compact set $[a + \delta, b]$ is covered by the open intervals $\{(a_j, b_j + \delta_j)\}_{j=1}^\infty$, so there exists a finite subcover $\{(a_j, b_j)\}_{j=1}^N$. Without loss of generality, let $a_1 < a_2 < \dots < a_N$. By construction $b_j + \delta_j \in (a_{j+1}, b_{j+1} + \delta_{j+1})$. We then get:

$$\begin{aligned}
 \mu_0(I) &= F(b) - F(a) \\
 &\leq F(b) - F(a + \delta) + \epsilon \\
 &= F(b_N + \delta_N) - F(a_N) + \sum_{j=1}^{N-1} (F(a_{j+1}) - F(a_j)) + \epsilon \\
 &\leq F(b_N + \delta_N) - F(a_N) + \sum_{j=1}^{N-1} (F(b_j + \delta_j) - F(a_j)) + \epsilon \\
 &= \sum_{j=1}^N (F(b_j + \delta_j) - F(a_j)) + \epsilon \\
 &< \sum_{j=1}^N \left(F(b_j) + \frac{\epsilon}{2^j} - F(a_j) \right) + \epsilon \\
 &\leq \sum_{j=1}^N (F(b_j) - F(a_j)) + 2\epsilon \\
 &= \sum_{j=1}^N \mu_0(I_j) + 2\epsilon
 \end{aligned}$$

Taking $\epsilon \rightarrow 0$ finally shows that μ_0 is countable additive over disjoint unions of h -intervals. Thus, μ_0 is a well-defined premeasure defined on $\mathcal{A}$, the algebra of countable disjoint unions of h -intervals. $\square$

Having finally constructed the premeasure $(\mathbb{R}, \mathcal{A}, \mu_0)$, we can now apply Carathéodory's Extension Theorem.

Theorem 2.4.4. *Let F be a distribution function. There is a unique Borel measure μ_F on $\mathbb{R}$ such that $\mu_F((a, b]) = F(b) - F(a)$. If G is another such function, then $\mu_F = \mu_G$ if and only if $F - G$ is constant. Conversely, if μ is a Borel measure on $\mathbb{R}$ that is finite on all bounded Borel sets, define:*

$$F(x) = \begin{cases} \mu((0, x]), & x > 0, \\ 0, & x = 0, \\ \mu(-(x, 0]), & x < 0, \end{cases}$$

Then F is a distribution, and $\mu = \mu_F$.

Proof. Proposition 2.4.3 showed that each F induces a premeasure on $\mathcal{A}$, the collection of countable disjoint unions of h -intervals. It is clear that F and G induce the same premeasure if $F - G$ is constant, and that these premeasures are σ -finite. The assertion that there is a *unique* Borel measure comes from Carathéodory's Extension Theorem. Given the outer measure μ^* , induced by the premeasure on countable disjoint unions of h -intervals, define $\mu_F := \mu^*|_{\mathcal{M}(\mathcal{A})} = \mu^*|_{\mathcal{B}_{\mathbb{R}}}$. The converse direction follows from the monotonicity of μ . $\square$

There are many levels to this application of Theorem 2.3.2, Theorem 2.3.4, and Proposition 2.4.3. Different restrictions of the outer-measure space $(X, \mathcal{P}(X), \mu^*)$ give different σ -algebra to work in. We have the chain of σ -algebras:

$$\mathcal{A} \subset \mathcal{B}_{\mathbb{R}} \subsetneq \mathcal{M}^* \subset \mathcal{P}(X),$$

Where $\mu_F := \mu^*|_{\mathcal{M}(\mathcal{A})} = \mu^*|_{\mathcal{B}_{\mathbb{R}}}$ is the measure on $\mathcal{B}_{\mathbb{R}}$, and $\mu^*|_{\mathcal{M}^*}$ is the *complete* measure on $\mathcal{M}^*$, the set of all μ^* -measurable sets. One can rigorously show that the domain of $\mu^*|_{\mathcal{M}^*}$ is strictly larger than the domain of $\mu^*|_{\mathcal{B}_{\mathbb{R}}}$.

Definition 2.4.4. Let μ_0 be a premeasure on $\mathcal{A}$ (the algebra of countable disjoint unions of h -intervals) and F a distribution function on μ_0 . The restriction of the $(X, \mathcal{A}, \mu_0)$ induced outer measure onto all measurable sets is called the *Lebesgue-Stieltjes measure associated to F* .

For the remainder of this section, let μ denote the Lebesgue-Stieltjes measure on $\mathbb{R}$ associated to the distribution function F , and $\mathcal{M}_{\mu}$ the domain of μ .

Theorem 2.4.5. If $E \in \mathcal{M}_{\mu}$, then:

$$\begin{aligned} \mu(E) &= \inf\{\mu(U) \mid U \supset E \text{ and } U \text{ is open}\} \\ &= \sup\{\mu(K) \mid K \subset E \text{ and } K \text{ is compact}\} \end{aligned}$$

Proof.

□

Theorem 2.4.6. *** Let $F : \mathbb{R} \rightarrow \mathbb{R}$ be an increasing and right-continuous function. There exists a unique measure μ_F on $\mathcal{B}_{\mathbb{R}}$ such that $\mu((a, b]) = F(b) - F(a)$. for all a, b . If G is another such function, we have $\mu_F = \mu_G$ if and only if $F - G$ is constant. Conversely, if μ is a measure on $\mathcal{B}_{\mathbb{R}}$ that is finite on all bounded Borel sets adding we define:

$$F(x) = \begin{cases} \mu((0, x]), & x > 0 \\ 0, & x = 0 \\ -\mu((x, 0]), & x < 0, \end{cases}$$

then F is increasing, right continuous, and $\mu = \mu_F$.

Proof. 10/8/25

□

Note that we have the following chain of inclusions:

$$\mathcal{A} \subset \mathcal{B}_{\mathbb{R}} \subsetneq \mathcal{M}_{\mu} \subset \mathcal{P}(\mathbb{R})$$

This theorem gave us a measure μ^* on $\mathcal{M}_{\mu}$, but we only really every cared about measures on $\mathcal{B}_{\mathbb{R}}$ since they are generated by nice sets. We got a good bang for our buck. **Are we talking about sets on $\mathcal{B}_{\mathbb{R}}$ or sets on $\mathcal{M}_{\mu}$?** Okay so basically the book is saying that $\mu^*|_{\mathcal{M}(\mathcal{A})} := \mu_F \dots$ BUT.... we have the complete measure $\mu^* := \overline{\mu_F}$ on $\mathcal{M}^* := \mathcal{M}_{\mu} \dots$ Basically we are talking about properties that hold on $\mathcal{M}^* := \mathcal{M}_{\mu}$ (caratheodoyrys extension theorem gave us this $\mathcal{M}^*, \dots$) Ignore what I just said and look at this table:

$$\begin{aligned} \mathcal{A} &\leftrightarrow \mu_0 \\ \mathcal{B}_{\mathbb{R}} &\leftrightarrow \mu^*|_{\mathcal{M}(\mathcal{A})} \\ \mathcal{M}_{\mu} &\leftrightarrow \mu^*|_{\mathcal{M}_{\mu}} \\ \mathcal{P}(\mathbb{R}) &\leftrightarrow \mu^* \end{aligned}$$

We're gonna be proving properties on $\mathcal{M}_{\mu}$

Definition 2.4.5. A measure satisfying the property from Proposition ??

For the remainder of this section we will fix a complete Lebesgue-Stieltjes measure μ on $\mathbb{R}$ associated to the increasing, right continuous function F . We denote by $\mathcal{M}_{\mu}$ the domain of μ . **Also, the notation in Folland is horrible**

Theorem 2.4.7. Let $E \subset \mathcal{M}_{\mu}$.

- (1) $\mu(E) = \inf\{\mu(U) \mid U \supset E, U \text{ open}\}.$
- (2) $\mu(E) = \sup\{\mu(K) \mid K \subset E, K \text{ compact}\}.$

Proof.

□

Da Homework:

Exercise 2.4.1. Let $\mathcal{A} \subset \mathcal{P}(X)$ be an algebra, $\mathcal{A}_\sigma$ the collection of countable unions of sets in $\mathcal{A}$, and $\mathcal{A}_{\sigma\delta}$ the collection of countable intersections of sets in $\mathcal{A}_\sigma$. Let μ_0 be a premeasure on $\mathcal{A}$ and μ^* the induced outer measure.

- (a) For any $E \subset X$ and $\epsilon > 0$ there exists $A \in \mathcal{A}_\sigma$ with $E \subset A$ and $\mu^*(A) \leq \mu^*(E) + \epsilon$.
- (b) If $\mu^*(E) < \infty$, then E is μ^* -measurable if and only if there exists $B \in \mathcal{A}_{\sigma\delta}$ with $E \subset B$ and $\mu^*(B \setminus E) = 0$.
- (c) If μ_0 is σ -finite, the restriction $\mu^*(E) < \infty$ in (b) is superfluous.

Proof. (a) Let $E \subset X$ and $\epsilon > 0$. By the definition of $\mu^*(E)$, we can find $\{A_j\}_{j=1}^\infty \subset \mathcal{A}$ such that $\bigcup_{j=1}^\infty A_j \supset E$ and:

$$\begin{aligned}
 \mu^*(E) + \epsilon &\geq \sum_{j=1}^\infty \mu_0(A_j) && \text{Property of infimums} \\
 &= \sum_{j=1}^\infty \mu^*(A_j) && \text{Because } \mu^*|_{\mathcal{A}} = \mu_0 \\
 &\geq \mu^*\left(\bigcup_{j=1}^\infty A_j\right). && \text{Outer measures are subadditive}
 \end{aligned}$$

(b) $(\Rightarrow)$ Suppose E is μ^* -measurable. By (a), for every $n \in \mathbb{N}$, there exists $A_n \in \mathcal{A}_\sigma$ such that $E \subset A_n$ and $\mu^*(A_n) \leq \mu^*(E) + \frac{1}{n}$. Since $\mu^*(E) < \infty$, we have $\mu^*(A_n) - \mu^*(E) \leq \frac{1}{n}$. Define $B = \bigcap_{n=1}^\infty A_n \in \mathcal{A}_{\sigma\delta}$. Then for all $n \in \mathbb{N}$:

$$\begin{aligned}
 \mu^*(B \setminus E) &= \mu^*(B \cap E^c) \\
 &\leq \mu^*(A_n \cap E^c) && \text{Monotonicity \& } B \subset A_n \ \forall n \\
 &= \mu^*(A_n) - \mu^*(A_n \cap E) && \text{Since } E \text{ is } \mu^*\text{-measurable} \\
 &= \mu^*(A_n) - \mu^*(E) && \text{Since } E \subset A_n \\
 &\leq \frac{1}{n}
 \end{aligned}$$

Hence $\mu^*(B \setminus E) = 0$. $(\Leftarrow)$ Now let $B \in \mathcal{A}_{\sigma\delta}$ such that $E \subset B$ and $\mu^*(B \setminus E) = 0$. Apply part (a) to the set $B \setminus E$. For each $n \in \mathbb{N}$, there exists $A_n \in \mathcal{A}_\sigma$ such that $B \setminus E \subset A_n$ and $\mu^*(A_n) \leq \mu^*(B \setminus E) + \frac{1}{n} = \frac{1}{n}$. Since $\mathcal{A}_\sigma \subset \mathfrak{M}(\mathcal{A}) \subset \mathcal{M}^*$, and since $\mathcal{M}^*$ is a σ -algebra, the set $\bigcap_{n=1}^\infty A_n$ is measurable. Since $B \setminus E \subset A_n$ for every $n \in \mathbb{N}$, we have $B \setminus E \subset \bigcap_{n=1}^\infty A_n$. Since $\bigcap_{n=1}^\infty A_n \subset A_n$, for every $n \in \mathbb{N}$ we have $\mu^*\left(\bigcap_{n=1}^\infty A_n\right) \leq \mu^*(A_n) \leq \frac{1}{n}$. Hence $\mu^*\left(\bigcap_{n=1}^\infty A_n\right) = 0$. Since $(X, \mathcal{M}^*, \mu^*|_{\mathcal{M}^*})$ is a complete measure space, and since $\bigcap_{n=1}^\infty A_n$ is a null-set, every subset of $\bigcap_{n=1}^\infty A_n$ is measurable. In particular, $B \setminus E$ is measurable. Moreover, $B \in \mathcal{A}_{\sigma\delta} \subset \mathcal{M}^*$, hence B is also measurable. We can write $E = B \setminus (B \setminus E) = B \cap (B \setminus E)^c$. Note that $B \in \mathcal{M}^*$, and $(B \setminus E)^c \in \mathcal{M}^*$ since $B \setminus E \in \mathcal{M}^*$ and since $\mathcal{M}^*$ is closed under intersections (it is a σ -algebra), hence $E \in \mathcal{M}^*$ meaning that it is measurable.

(c) We did not use the restriction $\mu^*(E) < \infty$ in the $(\Leftarrow)$ direction, hence we only need to prove $(\Rightarrow)$ for when μ_0 is σ -finite. Let $X = \bigcup_{j=1}^\infty X_j$. Then $\mu_0(X_i) < \infty$. Define $E_n =$

$E \cap \left(\bigcup_{j=1}^n X_j \right)$. Then $\bigcup_{n=1}^{\infty} E_n = E$.

For every $n \in \mathbb{N}$, apply part (a) to E_n .

There exists $C_n \in \mathcal{A}_\delta$ such that $E_n \subset C_n$ and $\mu^*(C_n) \leq \mu^*(E_n) + \frac{1}{n}$. Define $C = \bigcap_{n=1}^{\infty} C_n$. Then $\mu^*(C) \leq \mu^*(C_n) \leq \mu^*(E_n) + \frac{1}{n}$. Rearranging gives $\mu^*(C) - \mu^*(E_n) \leq \frac{1}{n}$. But since $\mu^*(E_n) \leq \mu^*(E)$, we have $\mu^*(C) - \mu^*(E) \leq \mu^*(C) - \mu^*(E_n) \leq \frac{1}{n}$. Since this holds true for all $n \in \mathbb{N}$, we have $\mu^*(C \setminus E) = 0$. $\square$

Exercise 2.4.2. Let μ^* be an outer measure on X induced by the finite premeasure μ_0 . If $E \subset X$, define the *inner measure* of E to be $\mu_*(E) = \mu_0(X) - \mu^*(E^c)$. Show E is μ^* -measurable if and only if $\mu^*(E) = \mu_*(E)$.

Proof. ($\Rightarrow$) Suppose E is μ^* -measurable. Then E^c is also μ^* -measurable since $\mathcal{M}^*$ is a σ -algebra. We can write:

$$\mu^*(X) = \mu^*(E) + \mu^*(E^c).$$

Since $X \in \mathcal{A}$ and $\mu_0 = \mu^*|_{\mathcal{A}}$, we have:

$$\mu_0(X) = \mu^*(E) + \mu^*(E^c).$$

Subtracting both sides by $\mu^*(E^c)$ gives $\mu_*(E) = \mu^*(E)$.

($\Leftarrow$) Now suppose $\mu^*(E) = \mu_0(X) - \mu^*(E^c)$. Again, since $X \in \mathcal{A}$ and $\mu_0 = \mu^*|_{\mathcal{A}}$, we have:

$$\mu^*(E) + \mu^*(E^c) = \mu^*(X) = \mu_0(X) < \infty$$

For each $n \in \mathbb{N}$, Apply Exercise 1, part (a) to E . Then there exists $A_n \in \mathcal{A}_\sigma$ such that $E \subset A_n$ and $\mu^*(A_n) \leq \mu^*(E) + \frac{1}{n}$. Define $B := \bigcap_{n=1}^{\infty} A_n$. Note that B is measurable since $B \in \mathcal{A}_{\sigma\delta} \subset \mathcal{M}^*$. Moreover since $E \subset A_n$ for all $n \in \mathbb{N}$, we have $E \subset B$. Together:

$$\mu^*(E) \leq \mu^*(B) \leq \mu^*(A_n) \leq \mu^*(E) + \frac{1}{n}.$$

Since this holds true for all $n \in \mathbb{N}$, it must be that $\mu^*(E) = \mu^*(B)$. Quickly note:

$$\begin{aligned} \mu^*(B) + \mu^*(B^c) &= \mu^*(X) \\ &= \mu^*(E) + \mu^*(E^c). \end{aligned}$$

Since $\mu^*(B) = \mu^*(E) < \infty$, we have $\mu^*(E^c) = \mu^*(B^c)$. Now by the measurability of B :

$$\begin{aligned} \mu^*(E) + \mu^*(E^c) &= \mu^*(E) + \mu^*(E^c \cap B) + \mu^*(E^c \cap B^c) \\ &= \mu^*(E) + \mu^*(E^c \cap B) + \mu^*(B^c) \\ &= \mu^*(E) + \mu^*(E^c \cap B) + \mu^*(E^c) \end{aligned}$$

Hence $\mu^*(E^c \cap B) = \mu^*(B \setminus E) = 0$. By Exercise 1, part (b), E is μ^* -measurable. $\square$